

Comprehensive survey of UAVs communication networks

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ABSTRACT

Recently, technologies related to Unmanned Aerial Vehicle (UAV) are growing rapidly particularly sensors, networking, and processing technologies. Accordingly, governments and industry have heavily invested in the studies of UAVs and improving their performances for reliable and secure deployments. The design methods and the investigation of UAVs systems have progressed from mono-UAV uses to multi-UAVs and cooperative UAVs systems that need a high level of coordination and collaboration to perform tasks which require new networking models, approaches, and mechanisms for highly mobile nodes involving many complex parameters and constraints. In this context, this paper provides more details and offers a thorough investigation concerning UAV communication protocols, networking systems, architectures, and applications. In addition, we discuss UAV solutions as well as highlighting important technical challenges and open research issues requiring further studies and R&D work.

1. Introduction

Rapid progress in the design and the improvement of Unmanned Aerial Vehicles (UAVs) of various sizes, shapes, and their communication capabilities have taken place recently. UAVs can fly autonomously by attached microprocessors or can be operated distantly without taking any human personnel. Because of their flexibility, easy installation, low maintenance costs, versatility, and relatively small operating cost, the use of UAVs provides new ways for commercial, military, civilian, agricultural, and environmental applications such as border surveillance [1,2], relay for ad hoc networks [3,4], managing wildfire [5], disaster monitoring [6], wind estimation [7], traffic monitoring [8,9], remote sensing [10], and search and destroy operations [11].

1.1. UAV system overview

Some of these applications require a single UAV system and others like area monitoring for hazardous environments require multi-UAVs systems. Although single-UAV systems are used for decades, by operating and developing one large UAV, exploiting a set of small UAVs has many advantages. In the single UAV systems, each UAV act as an isolated node, it can only communicate with the ground node. Thus, the UAV communication system is based only on UAV-to-infrastructure communication, and the communication between the UAVs can be established through the infrastructure. The capability of a single UAV

system is limited compared to the multi UAV system which has many advantages. Firstly, missions are generally completed at a lower cost with multi UAV systems. Besides, the collaborative work of UAVs can improve the performance of the system. Moreover, in a multi UAV system, if the UAV fails in a mission, the operation can survive with the other UAVs, and missions are generally completed more quickly and efficiently with multi-UAV systems.

For successful and efficient mission completion multiple UAVs can be utilized because of their capabilities, flight time, and limited payload [12]. Communication and networking are necessary to enable co-operation, organize multiple UAVs, and achieve autonomous UAVs networks. Ad hoc networks between multiple UAVs can be an alternative communication solution. In the ad hoc UAVs network, only some UAVs are connected to the ground base, but all UAVs form an ad hoc network. In this system, UAVs can communicate with other UAVs and the ground base. Ad hoc UAVs networks can be considered as a special structure of Mobile Ad-hoc Network (MANET), and Vehicular Ad-hoc Network (VANET). Whereas, UAVs networks have some special characteristics compared to the existing Ad hoc networks. Nodes in the UAVs networks are characterized by their high mobility degree. VANET and MANET nodes are cars and walking men respectively, UAVs fly in the sky. The high mobility of UAVs influence the network topology which changes more frequently compared to the topology of the MANET or the VANET. Moreover, MANET and VANET aim to create peer-to-peer connections. UAVs networks also necessitate peer-to-peer connections to ensure coordination and collaboration between UAVs. In

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the most of the time, UAVs collect data and relay it to the Ground Control Station(GCS). Therefore, it is required to ensure the UAV-to-UAV (U2U) communication and the UAV-to-Infrastructure (U2I) communication. Consequently, UAVs networks must ensure peer-to-peer communication and converge cast traffic at the same time. Furthermore, distances between UAVs are much longer than nodes in the MANETs and VANETs. Therefore, in order to create reliable communication links between UAVs, it is required to increase their communication range.

1.2. UAVs system challenges and research motivations

The most prominent design problem of multi-UAV systems is the communication which is important for coordination and collaboration between the UAVs. UAVs can be used in aerial sensor networks which is composed of multiple data sources distributed in different areas where UAVs nodes are used to collect information. It may include different types of sensors, and each sensor may require different data delivery strategies. If there is a need to use different sensors, they should be loaded on different UAVs, e.g., one UAV can be loaded with an infrared camera, while another UAV is equipped with a high-resolution camera.

On the other hand, UAVs networks have many challenging parameters [1,13] such as limited bandwidth, high mobility, intermittent connectivity, bounded transmission range, and unpredictable noisy channels. These challenges produce different issues in the ad hoc multi-hop environment such as collisions, and transmission delays [14]. For instance, it is difficult to keep the transmission range between two UAVs moving in opposite directions with very high velocity.

Due to this UAV challenging issues, more investigations and deep studies related to UAV communication systems are required. Accordingly, among the objectives of this survey is to identify the challenges and design characteristics and constraints of the UAVs networks. Moreover, we investigate the essential needs and functions for communication in UAV-based systems, and we propose various architectures that can be used for multi-UAV systems. We study the UAVs related protocols and mechanisms.

1.3. Paper organization

All acronyms used throughout this survey paper are given in Table 1. The paper is organized as presented in Fig. 1. In Section 2, we provide a comparative study of the existing surveys. We present several UAV's systems and architectures in Section 3. Section 4 highlights UAV networking technologies towards global connectivity. UAVs systems considerations are presented in Section 5. In Section 6, we provide a review of the existing UAV's communication protocols and mechanisms. In Section 7, we present UAV mobility models. In Section 8, we present security challenges of UAV networks. UAV localization methods are highlighted in Section 9. In Section 10, we highlight some QoS provisioning for UAV networks. In Section 11 we present current research related to UAVs applications. Furthermore, we identify and we present the existing multi-UAV projects, testbeds and simulation environments in Section 12. We identify open research issues in Section 13. Finally, we conclude the paper in Section 14.

2. Related works

Nowadays, there exist many studies and investigations related to UAVs networks. Accordingly, many UAVs surveys have been elaborated, in this context, this section overviews recent existing surveys. They mainly provided a general overview of the UAVs communication models or exhibited the related applications, characteristics, challenges, and open issues, or gave some solutions to respond to a particular requirement (likes security issues, Medium Access Control (MAC) protocols, quality of service, routing protocols, etc.). To the best of our knowledge, there is no previous work which studied all these topics in

the same paper.

Authors in [15] overview the important issues in UAV communication networks. They studied the possibilities of the communication architecture and they assumed that the meshed communication architecture is the better in terms of flexibility, reliability, and performance. Furthermore, they studied the applicability of such routing protocols to UAV networks and they considered 5 categories of routing protocols for UAV networks: Static protocols, Proactive protocols, On-demand or Reactive protocols, Hybrid protocols and Geographic 2-dimension and 3-dimension protocols. Moreover, the authors proposed an additional classification of routing protocols for Delay Tolerant Networks (DTN). They consider three types: deterministic routing, stochastic routing, and social networks based approach. The stochastic routing algorithms are used when the network behavior is random and unknown. Moreover, the authors studied the significant effect of handovers in the network communication. Thus, handovers allow the continuity of the network communication with increasing negligible message latency. They discussed the applicability of classics protocols for seamless handover in UAV networks based on several considerations like security, packet loss rate, latency, node mobility, and speed. Furthermore, they discussed energy efficiency in UAV networks at several layers such as the network layer, the data link layer, physical layer. Moreover, the authors sketched energy conservation through cross-layer protocols.

In [16], the authors studied ad hoc UAVs network which is named Flying Ad-Hoc Networks (FANETs). They considered different FANET application scenarios and several FANET designs [17,18] for extending the scalability of multi-UAV applications. The authors focused on the UAV swarms scenario [19] using small UAVs. Thus, a swarm of UAVs requires coordinated functions and must prevent themselves from collisions to complete their operation successfully. Besides, the authors discussed the most prominent FANET design considerations which are adaptability, scalability, latency, UAV platform constraints and bandwidth requirement. Researchers studied the applicability of communication protocols to FANETs at the physical layer, the MAC layer, the network layer, the transport layer and their cross-layer interactions and presented the open research issues of them. The authors supposed that the success of FANET designs is joined to the reliability of the communication architecture and the reliable transport mechanism. Researchers highlight some researches [20] which enhance interoperability between nodes. Equally, several cross-layer architectures [21,22] are developed to resolute the performance problems of the wireless environment. The authors studied the existing FANET testbeds and simulators which are explored to motivate new FANET researchers to design solutions and to test the communication approaches and protocols with different mobility patterns.

In [23], the authors identified the concepts in UAVs networks. Besides, they cited the advantages of using multi UAVs systems. Researchers classified routing protocols in four essential categories: static protocols, proactive protocols, reactive protocols and hybrid protocols. The authors reported that the existing routing protocols are not efficient for UAVs networks communication because they are not fault-tolerant and scalable, and they are not developed to ensure the communication between UAVs and the ground stations.

The aforementioned surveys didn't discuss the coverage problem of UAVs. In [24], the authors studied the area coverage problem in the UAV network. The most important challenges are the coverage ability, the mobility of UAVs, the energy constraints and lifetime, the connect strategy and the obstacles. The authors compared some researches and they discussed the considerations and the limitations on the basis of the mentioned factors. Moreover, the authors considered two categories of coverage: static coverage and dynamic coverage depending on the UAVs motions. Equally, coverage problems are divided into homogeneous coverage and heterogeneous coverage which is more complicated.

After surveying the literature, we have identified many research works that survey the applications of UAVs for traffic management. In

Table 1
Acronyms used in this paper.

Acronym	Text	Acronym	Text
AMUAV	Adaptive MAC UAV	AoA	Angle of Arrival
AODV	Ad-hoc On-demand Distance Vector	A2G	Air to Ground
BATMAN	Better Approach to Mobile AdHoc Network	BC	BlockChain
BIA	Backward Information Ant	BICSF	Bio-Inspired Clustering Scheme
BIR-SLB	Bio-Inspired Routing with Shortest path and Load-Balancing	BS	Base Station
BTS	Busy to Send	CDMA	Code Division Multiple Access
CF-MAC	Collision-Free MAC	CPS	Cyber Physical System
CSI	Channel State Information	CSM	Collision Solver Manager
CURS	Closest UAV Relay Selection	DAG	Directed Acyclic Graph
DCR	Data-Centric Routing	DDR	Distributed Dynamic Routing
DPTR	Distributed Priority Tree-Based Routing	DSDV	Destination-Sequenced Distance Vector
DSSS	Direct Sequence Spread Spectrum	DSR	Dynamic Source Routing
DTN	Delay Tolerant Networks	EHSR	Extended Hierarchical State Routing
FANET	Flying Ad-Hoc Networks	FEA	Forward Exploration Ant
FFT	Fast Fourier Transform	FGQPA	Fountain-code based Greedy Queue and Posit
FPGA	Field Programmable Gate Arrays	GCS	Ground Control Station
GDSTR-3D	Greedy Distributed Spanning Tree Routing-3D	GHG	Greedy-Hull-Greedy
GLSR	Geographic Load Share Routing	GPMOR	Geographic Position Mobility Oriented Routing
GPS	Global Positioning System	GPSR	Greedy Perimeter Stateless Routing
GRASP	General Robotics Automation Sensing and Perception	GRG	Greedy-Random-Greedy
HAPS	High Altitude Platforms Station	HGA	Hybrid Genetic Algorithm
HRP	Hybrid Routing Protocols	IMU	Inertial Measurement Unit
INS	Inertial Navigation System	IoT	Internet of Things
LCAD	Load Carry and Deliver Routing	LiDAR	Light Detection and Ranging Sensor
LOS	Line Of Sight	LQ-OLSR	Link-Quality OLSR
LSP	Link State Packet	MAC	Medium Access Control
MANET	Mobile Ad-hoc Network	MAVLink	Micro Air Vehicle Link
MDT	Multi-hop Delaunay Triangulation	MIM	Man In the Middle
MLHR	Multi-Level Hierarchical Routing	ML-OLSR	Mobility and Load OLSR
MMSE	Minimum Mean-Square-Error	MPR	Multi-Point Relay
MURS	Maximum UAV Relay Selection	NOMA	Non-Orthogonal Multiple Access
OLSR	Optimized Link State Routing Protocol	OMA	Orthogonal Multiple Access
POI	Points of Interest	PRP	Proactive Routing Protocols
PUDT	Partial Unit Delaunay Triangulation	QoS	Quality of Service
RAVEN	Real-time indoor Autonomous Vehicle Environment	RD	Random Direction
RGR	Reactive Greedy Reactive	RRP	Reactive Routing Protocols
RREQ	Route Request	RREP	Route Response
RSSI	Received Signal Strength Indicator	RW	Random Walk
RWP	Random Waypoint	SAODV	Time-Slotted On-demand Distance Vector
SAR	Search and Rescue	SC	Superposition Coding
SIC	Successive Interference Cancellation	SINR	Signal Interference Noise Ratio
SNR	Signal-to-Noise Ratio	SRCM	Semi-Random Circular Movement
ST	Smooth Turn	SUAAVE	Sensing Unmanned Autonomous Aerial Vehicles
SUAP	Secure UAV Ad-Hoc Routing Protocol	TDMA	Time-Division Multiple Access
TDNT	Time-Dependent Network Topology	TENSR	Tactical Edge Network Social Routing
TRN	Terrain Referenced Navigation	UAV	Unmanned Aerial Vehicle
UE	User Equipment	URS	UAV Relay Selection
USMP	UAV Search Mission Protocol	UVAR	UAV-Assisted VANET Routing Protocol
UURS	Uniform UAV Relay Selection	U2I	UAV-to-Infrastructure
U2U	UAV-to-UAV	VANET	Vehicular Ad-hoc Network
VO	Visual Odometry	WSN	Wireless Sensor Network
ZHLS	Zone-based Hierarchical Link State routing	ZRP	Zone Routing Protocol

[12], the authors studied the applications of distributed processing systems for UAV/drones. They classified these applications into categories depending on their purpose. The authors considered applications that employ cooperative UAVs for object detection and applications using multiple UAVs for tracking targets and sharing the gathered data from data sources with each other. Besides, the authors focused on applications for surveillance [25,26] and applications using sensors for data collection [27] using Wireless Sensor Network (WSN). They studied algorithms which must be used in the cooperative swarms of UAVs such as in the path planning, the navigation, the collision avoidance, the coordination, and the monitoring.

In [28] the authors provided a technical report of researches in the area of UAVs traffic monitoring and management and they studied their advantages and their disadvantages.

In [29] researchers proposed a taxonomy for UAVs networks based on their network formation. They studied the single UAVs and the multi-UAVs network formations, and they provided several applications

for each system. The multi-UAVs systems are categorized based on the distribution parameters used by the UAV system. They considered eight categories: the cooperative UAVs networks, the opportunistic relaying UAVs networks, the task allocations UAVs networks, the UAV-aided networks, the Delay-Tolerant UAVs networks, the cluster formations UAVs networks, the self-organizing UAVs networks, and the heterogeneous UAVs networks. Furthermore, the authors provided various software solutions, simulators, and testbeds and they discussed various open issues.

In [30] the authors studied the UAV networks characteristics and requirements for civil applications. Besides, they classified civil UAV applications from a communication viewpoint into four categories: search and rescue, construction, coverage, and delivery of goods. Then, the authors detailed the requirements and the challenges of applications based on Quality of Service(QoS) communication and applications based on quantitative communication. Likewise, they discussed the feasibility of the existing wireless technologies in the UAVs networks.

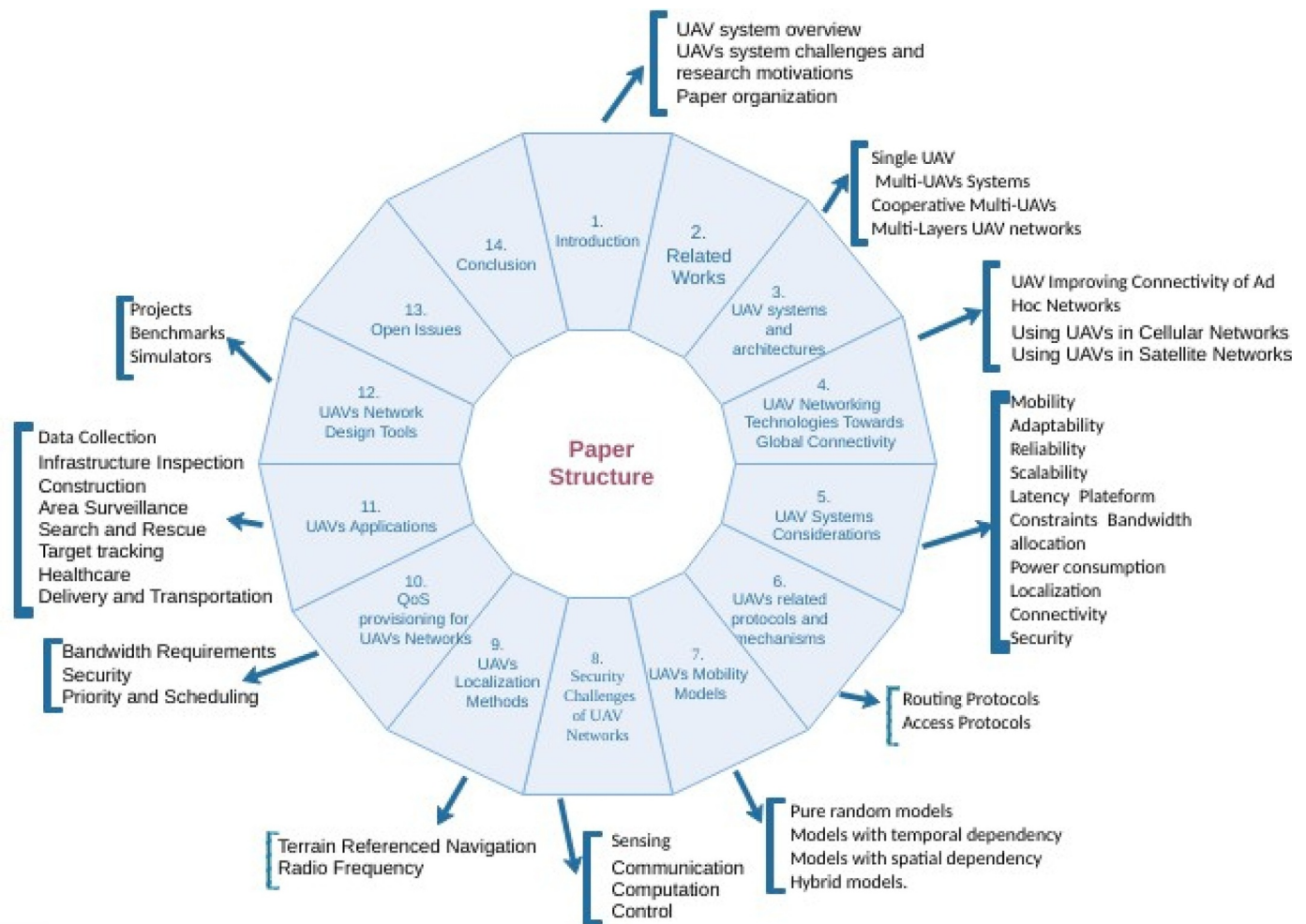


Fig. 1. Survey taxonomy.

Moreover, the researchers reported some open issues and research challenges.

In [31] the authors studied the UAVs communications security requirements. Besides, they detailed two specific attacks (Jamming and Spoofing of the Global Positioning System (GPS), the WiFi attacks) and they described the defense solutions.

In [32] the authors reported the basic networking architectures. They provided the characteristics of the different types of channels: the UAV-ground channel and the UAV-UAV channel. Furthermore, they discussed the UAV communication design considerations. They treated the UAV controlled mobility and they discussed two techniques to improve the performance of the UAVs communications: the UAV-enabled mobile relaying and the device-to-device enhanced UAV information dissemination.

In [33] the authors proposed a review of the communication and networking of UAV-based systems. They studied functions, services, and requirements for the UAV system and they provide a classification for the associated architectures. In addition, the paper presented middleware layer services used to ensure seamless communication and maintain heterogeneous network interfaces. Besides, they discussed the use of UAVs in collecting data from WSNs.

In [34] the researchers presented the main communication features of UAVs networks. They provided the main categories of wireless communications in UAV networks and several wireless technologies that can be used. They reviewed also the main evaluation tools to assess UAVs networks. Further, the authors indicated the main open challenges for the design and the evaluation of UAV networks.

The authors of [35] presented an overview of the FANET architectures. They introduced major routing techniques and their features. Besides, they proposed a novel taxonomy for the routing protocols for eight categories which are Topology-based, Secure-based, Swarm-based, Hierarchical-based, Energy-based, Heterogeneous-based, Position-based, and DTN-based. They also discussed some challenging open issues.

In [36], the authors presented background information about FANET. They treated FANET communication issues. They studied the mobility models and the trajectory optimization. They considered some security issues.

The authors of [37] gave the taxonomy of the wireless networks. They provided the UAV networks characteristics and related design issues. They introduced a lot of routing protocols and they studied QoS aware routing protocols and QoS metrics for security.

The authors of [38] provided a taxonomy for routing schemes in FANETs and they gave a deep comparison of these routing schemes. Besides, they presented some open issues.

The aforementioned research works identified several problems in UAVs network and they studied existing approaches and different solutions and protocols. Table 2 presents and compares these research works based on their studied points. Moreover, Table 3 compares these research works based on the provided potential UAVs' applications. Whereas, to the best of our knowledge, there is no survey that includes all the studied points. Based on all these research works, we provide in this paper a deep study on the concepts and challenges in the UAVs network and we give a holistic view of the UAV communication system.

Table 2
Related surveys studied points.

Studied Points	[16]	[18]	[28]	[24]	[23]	[12]	[29]	[30]	[31]	[32]	[33]	[34]	[35]	[36]	[37]	[38]	Our Survey
Characteristics	✓	✓		✓	✓	✓	✓	✓		✓			✓	✓	✓	✓	✓
Considerations	✓	✓		✓	✓	✓	✓	✓		✓					✓	✓	✓
Applications		✓	✓	✓		✓	✓	✓				✓	✓	✓	✓		✓
Architectures	✓	✓		✓	✓		✓			✓	✓	✓	✓	✓			✓
Physic Layer	✓										✓						✓
Link Layer	✓										✓	✓		✓			✓
Network Layer	✓	✓		✓	✓						✓		✓	✓	✓	✓	✓
Transport Layer	✓																✓
Seamless Handover	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							✓
Channel Security		✓	✓	✓	✓	✓	✓	✓	✓	✓			✓	✓	✓		✓
Energy Management/ Power Control	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Projects/Simulators	✓		✓				✓	✓	✓			✓	✓	✓	✓	✓	✓
Open Issues and Future Challenges	✓		✓		✓		✓	✓			✓	✓	✓	✓	✓	✓	✓

3. UAV systems and architectures

In this section, we provide a deep review of UAVs networking architecture. Basically, we focus on multi-UAVs systems.

3.1. Single UAV systems

The single UAV systems are used for navigation and control for monitoring distant targets especially in unknown geographical territories and flight conditions to prevent any loss of pilots. A single UAV system does not mean that there is only one UAV acting in the network, but the network may have many UAVs. In this system, each UAV act as an isolated node as shown in Fig. 2. The ground node is at the center and the UAVs can directly communicate with the ground node. Thus, the UAV communication system is based only on UAV-to-infrastructure communication.

The communication between the UAVs can be established but through the infrastructure which lets the capability of a single UAV system is limited. In [39] the authors presented a framework for autonomous flying using the AR.Drone with minimal hardware and software additions. The system performs all the sensing and computations on-board, making the system independent of any base station or remote control. High-level navigation, computer vision, and control tasks are carried out in an external processing unit that steers the vehicle to the desired location.

In [40] the researchers presented efficient road detection and tracking framework in UAV videos. They used a graph detection approach to accurately extract a specified road region during the initialization stage and in the middle of the tracking process. The authors developed a fast homograph road tracking scheme to automatically track road areas.

In [41] the authors proposed a fast target localization method basing on multi-point observation using a single UAV system. Along the expected trajectory, the UAV is flying to give multi-point target

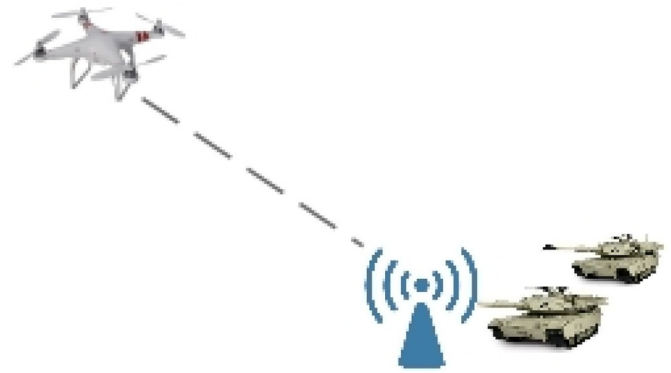


Fig. 2. Single UAV system .

observations to estimate the target location. To predict the target height, the authors used the steepest descent method assuming that the area of the particular region is minimal. If the height information is determined, then the exaction of the target position is computed.

In [42] the authors studied the performance of the system using a single UAV to monitor a specific size of an area for tracking multiple objects. They assumed that the objects are moving according to a linear velocity model with noise. They derived a performance bound for the variation of the estimated error position of objects. This performance bound will be used to calculate the required visitation period of an object, such that the probabilities of the error in the position estimate.

3.2. Multi UAVs systems

Today, most public and civil applications use multi UAV systems [43]. In such system, the UAVs are smaller and less expensive, and work in a coordinated manner. The most prominent design problem for multi-UAV systems is the communication which is important for

Table 3
Comparison of related surveys in terms of applications .

Applications	[16]	[18]	[28]	[24]	[23]	[12]	[29]	[30]	[31]	[32]	[33]	[34]	[35]	[36]	[37]	[38]	Our Survey
Data Collection	✓				✓					✓	✓	✓					✓
Infrastructure Inspection																	✓
Construction								✓			✓			✓			✓
Area Surveillance	✓	✓			✓	✓					✓				✓		✓
Search and Rescue								✓			✓			✓	✓		✓
Target Tracking	✓	✓			✓	✓											✓
Healthcare																	✓
Delivery/Transportation		✓						✓			✓			✓			✓

coordination and collaboration between the UAVs. If all UAVs are directly connected to an infrastructure, the communication between UAVs can be realized through the infrastructure. Thus, if a UAV is outside of the infrastructure coverage due to the dynamic environmental conditions, node movements or area structures, it becomes disconnected. Architecture based on only infrastructure communication restricts the capabilities of the multi-UAV systems. Consequently, the UAV communication system must be based on both UAV-to-infrastructure and UAV-to-UAV communications to improve the network reliability. The advantages of the multi-UAV systems can be summarized as follows:

Cost: The missions are completed at a lower cost with a multi-UAVs system compared to a single UAV system [44].

Scalability: Collaborative work of UAVs can improve the performance of the system in comparison to a single-UAV system. The multi-UAVs systems can provide wide-area monitoring and multi-objects tracking. Therefore, multi-UAVs systems are scalable compared to single-UAV systems.

Survivability: In a multi UAV system, if the UAV fails in a mission, the operation can survive with the other UAVs.

Speed-up: Missions are generally completed more quickly and efficiently with multi-UAVs systems as compared to a single UAV system [45].

In multi-UAV systems, topologies that can be established are a star, multi-star, mesh and hierarchical mesh. In the case of the star topology network, one ground node is at the center of the star and the UAVs can directly communicate with the ground node. Thus, the UAV communication system is based only on UAV-to-infrastructure communication.

In the case of the multi-star star topology network as illustrated in Fig. 3, the UAVs must form multiple stars and one node from each group connects to the ground station. Star configurations suffer from high latency because the downlink length is longer than inter-UAV distance and all communication must pass through the GCS. Also, if there is no communication with GCS, the mission fails because there is no inter-UAV communication. Moreover, it is required to have more expensive bandwidth downlinks.

In the case of the mesh topology network, the UAVs are interconnected and only one UAV may connect to the control center as illustrated in Fig. 4.

In the case of the hierarchical mesh topology network, the UAVs must form multiple mesh networks and one node from each group connects to the other groups. Also, a small number of UAVs may connect to the control center as illustrated in Fig. 5.

In a mesh topology network, nodes are interconnected and packets can pass through intermediate nodes and find its way from any source to any destination in multiple hops. Mesh networks are flexible, reliable and offer better performance characteristics than star networks. Furthermore, mesh networks are more suitable for UAVs network because they can support the self-forming and the reorganization features.

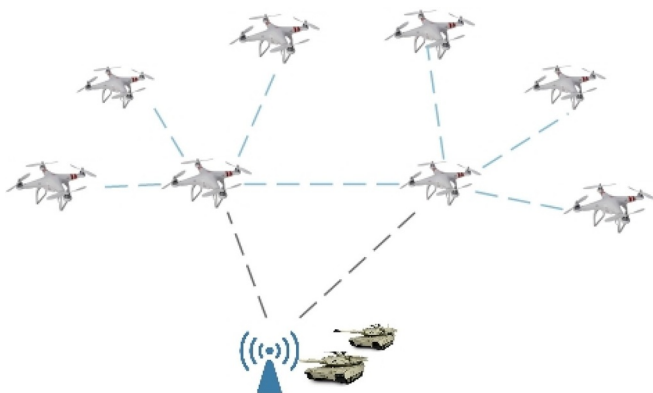


Fig. 3. Multi-star configuration system.

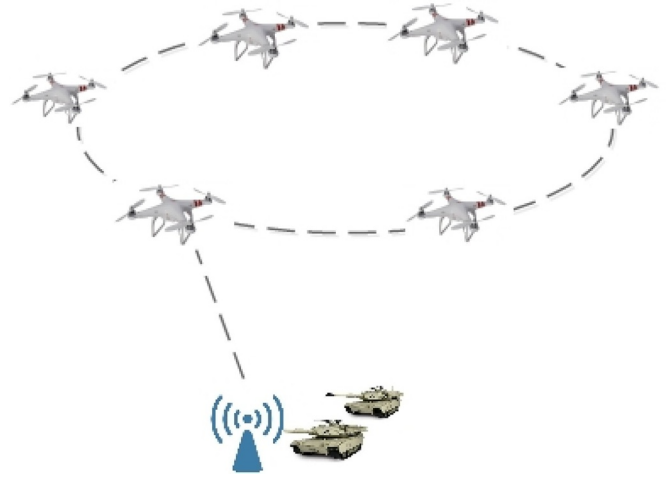


Fig. 4. Mesh configuration system .

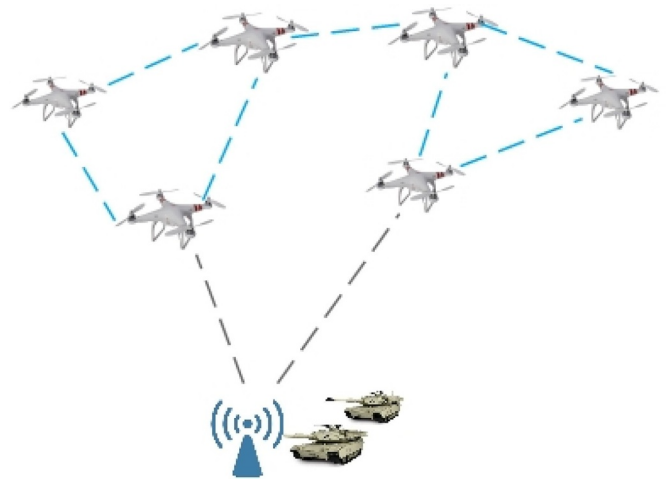


Fig. 5. Hierarchical mesh configuration system .

Thus, when a node fails, the rest of the nodes reconfigure the network among them.

In the rest of this section, the basis for the usage of multi-UAVs for various operations and applications are discussed. In fact, we classified the UAVs architectures into two categories which are cooperative Multi-UAVs and multi-layers UAV networks as shown in Fig. 6.

3.3. Cooperative Multi-UAVs

Cooperative Multi-UAVs architecture is used in a mission scenarios involving several UAVs with different communication characteristics. Cooperative Multi-UAVs architecture includes cooperative UAVs based task achievement, opportunistic relaying networks, and delay-tolerant UAVs networks.

3.3.1. Cooperative UAVs based task achievement

A cooperative network is a specialized graph in which each node acts with respect to the role of another node. The cooperative UAVs based task achievement execute complex tasks with coordinated operability. A cooperative UAV network can be static or dynamic. In the static network, the functionalities are predefined. Whereas in the dynamic network, the node behavior is not known and the network topology may change which affects its performance.

In [46] the authors discussed the complexity of multi-UAVs cooperative control. The authors discussed several issues of UAVs cooperative features. They decomposed the hierarchical architecture of

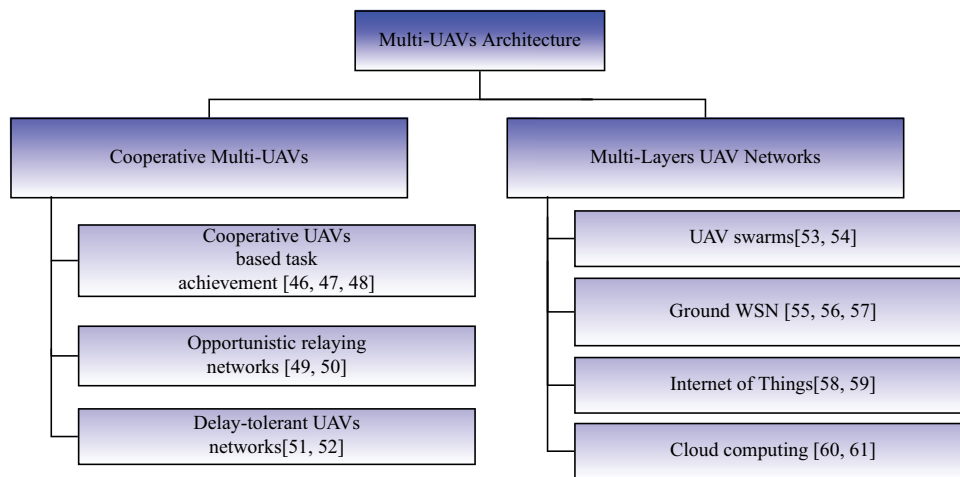


Fig. 6. Classification of Multi-UAVs architectures.

UAVs into various graph models, enumerations and cooperative assignments to easily understand the autonomous vehicles connections. In [47] the researchers developed a cooperative ad-hoc network between the UAV nodes and the ground nodes. The guidance map support is provided to the network nodes. Cooperative networks can generate estimation networks that are formed by a combination of nodes over wireless links to carry out tasks of virtual computations. These cooperative networks use the estimations and the approximations to complete complex tasks.

In [48] the authors generated the optimal trajectory planning of cooperative UAVs for simultaneous operations. They used intelligent and dynamic programming concepts. Researchers proposed a conditional probabilistic approach based on a number of targets to create the decision model.

3.3.2. Opportunistic relaying networks

UAVs networks are characterized by frequent link failure, so, the opportunistic relaying is required. This allows enhanced utilization of network parameters and resources. In [49] authors proposed an algorithm for opportunistic relaying for a special category of the temporary network. The framework is based on the formation of controllers, helper modules, and resource reservoirs and a useful utility programs. In [50] authors formed a virtual environment for tracking nodes in the UAVs network, they consider the resource relaying for opportunistic networks. Researchers evaluate their work, using the OPNET simulator, based on the number of times the target is detected and the time for detection of these nodes. The approach is founded on cooperative resource utilization with the help of an intelligent detection system.

3.3.3. Delay-tolerant UAVs networks

Delay-tolerant UAVs networks are characterized by low connectivity, they use a store-and-forward approach for the routing protocol because the links are not continuously available. These networks have similar challenges as traditional networks.

In [51] authors used UAVs as DTN relays to enable the communication between the ground nodes. Each UAV hovers over its home-ground node until it is triggered to transmit messages to other ground nodes. It uses a genetic algorithm to determine the route that performs the most (timely) deliveries.

In [52] authors considered delay-tolerant UAVs network. They used spray and wait approach for path selection and management. Whereas, due to the lack of end-to-end connectivity, the continuous link fracture, and the high latency, the overheads during path selection increase. The proposed approach does not perform well.

3.4. Multi-layers UAV networks

Multi-layers UAV network architectures are based on UAVs network that cooperates with other layers such as Internet of Things(IoT) or WSN network or with cloud computing systems.

3.4.1. UAV swarms

UAVs can be organized in different swarms when there are many UAVs used in the network and the mission area is large. The swarm behavior can complete complex missions successfully and swarm UAVs must prevent themselves from collisions. Whereas it requires coordinated functions and UAVs must communicate with each other to achieve the coordination. To improve the UAVs payload capacity, some application uses the model of dividing UAVs into clusters to operate in different mission areas. In each cluster, there is a node (cluster Head) which is in connection with the upper/lower layers (ground stations, UAVs, satellites, etc.) directly or indirectly. It is responsible to assure the connectivity between UAVs in the same cluster. Cluster formations can provide an efficient approach for networks that have a higher failure rate. In [53] researchers proposed a swarm formation using multiple UAVs for realizing complex tasks. The authors perform flocking analysis for the cooperative mode to evaluate the performance of their approach. In [54] authors proposed a framework for UAVs swarms. The framework improves the sensing capabilities of the UAVs. It considers simple parallel search and forwards moving parallel sweep. The algorithm operates between upper and lower bounds that are computed on the basis of UAV failure probability.

3.4.2. Ground WSN

The aerial sensor network is composed of multiple data sources distributed in different areas and UAV nodes used to collect information. Thus, it moves around the environment, senses with the sensors and relays the collected data to the GCS. They usually register environmental values such as temperature, air pressure, wind, etc. If there is a need to use different sensors, they could be loaded on different UAVs, e.g., one UAV can be loaded with an infrared camera, while another UAV is equipped with a high-resolution camera. Furthermore, to achieve a specific task, we need aerial sensors and ground sensors. Accordingly, the sensing network will be structured into two layers: UAVs layer (including Aerial sensing layer) and ground WSN.

In [55] authors proposed a framework for heterogeneous UAVs to detect fire. The framework integrates infrared sensors and data fusion by obtaining values from computer-aided modules. The proposed framework localizes fire detection based on the position of UAVs and the

localization of cameras. In [56] authors proposed a WSN algorithm for data collection using groups of UAVs. The communication architecture relies only on ad-hoc UAVs, sensors and the GS. Researchers proposed a distributed algorithm to maintain the cyclic examination of a group of Points of Interest (POI). The proposed algorithm takes into account the multiple data sources distributed and reinforcements that may occur. In [57] authors studied the opportunistic routing (OR) for UAV assisted WSNs. Further, they introduced two new OR protocols. The first is ANOR protocol which allows the source node to share its traffic to the neighbors which are in its range. The second protocol is HVOR where the source node transmits packets to a single UAV having the highest speed. Therefore, HVOR dynamically selects a route and finds which sensor will be the forwarder.

3.4.3. Internet of Things (IoT)

UAVs can play an important role in IoT vision and can offer several IoT value-added services. In [58] authors provided a conceptual architecture for the Internet of Drone (IoD). The architecture covers three main networks which are the air traffic control network, the cellular network, and the Internet. In fact, the layered architecture provides services for several UAV applications such as surveillance, search and rescue, etc. In [59] authors investigated efficient UAV deployment and mobility to collect data from ground IoT devices. The framework optimizes the three-dimensional (3D) UAVs placement and mobility and it enables reliable uplink transmissions for IoT devices using a lower total transmit power.

3.4.4. Cloud computing

UAVs are characterized by computing and storage resources limitations, that's why intensive computation cannot be performed on-board. To overcome this weakness cloud computing can be used to provide the storage and computation services. Researchers in [60] presented a mapping between UAVs resources and cloud computing resources. The framework provides both essential services such as mission organization service, ground station commands, broker service, and customized services including sensing services, image/data analysis and actuation services. The authors in [61] presented a service-oriented cloud-based on a UAV management system named Dronemap Planner which provides access to UAV through web services, schedule missions and ensures collaboration between drones. In fact, a cloud proxy server is developed acting as a moderator between the UAVs and the users. The communication between the UAVs, the users, and the cloud are based on the MAVLink protocol.

A comparative study of the aforementioned investigations related to network architectures is highlighted in Table 4.

Table 4
A comparative study of UAV networks architectures

Architectures		Advantages	Disadvantages
Cooperative Multi-UAVs	Cooperative UAVs based task achievement [46–48] Opportunistic relaying networks [49,50]	Execute complex tasks with coordinated operability Suitable for dynamic network Decrease link failures Enhance the utilization of network resources Improve the throughput by applying Opportunistic transmissions	Use the estimations and the approximations. UAVs collisions Complex computations Packet duplication High energy consumption
	Delay-tolerant UAVs networks [51,52]	Support UAVs with limited power resources Avoid the complexities of routing Improve the system management	Low connectivity High latency No consideration of the buffer and bandwidth capacity No consideration of the security
Multi-Layers UAV Networks	UAV swarms [53,54]	Many UAVs Used in the large mission area Few communications with the ground station	UAVs collisions Complex computations for coordination
	Ground WSN [55–57]	Multiple and distributed data sources Exploit UAVs sensors Fast data collect	Limited opportunities to communicate with sensors Limited UAV transmission range
	Internet of Things (IoT) [58,59]	Layered network control architecture Fast aerial packets delivery Various applications and services Improve traffic management	Low UAVs energy UAVs range limitations UAVs landing restrictions Congested urban airspace
	Cloud Computing [60,61]	Offload heavy computations from the UAVs to the cloud Provide computation and storage Services remotely	Latency in UAV communications UAVs range limitations The necessity to add security mechanisms

4. UAV networking technologies towards global connectivity

The use of UAVs is rapidly growing, they can admit several prospective applications and benefits in wireless systems. In this section, we present several potential applications for UAV networking technology towards global connectivity.

4.1. UAV improving connectivity of ad hoc networks

UAVs can expand the communication range and connectivity of ad hoc networks. They are used to maintain reliable communication in homogeneous and heterogeneous networks between transmitters and receivers which cannot directly communicate due to their long separation distance or obstacles. Transmitter and receiver nodes can be UAVs, vehicles, sensors, etc.

However, deploying UAV on such networks presents numerous technical challenges such as mobility management, connectivity, collaboration, energy efficiency, etc. To address these challenging problems, the researchers are in full swing to develop adequate solutions.

In this context, authors of [62] used UAVs to help to find survivors of disasters and provide network connectivity to the teams of rescuing. They proposed a distributed algorithm to discover targets in unknown locations and ensure the connectivity between the rescue teams in a multi-hop wireless network. Moreover, researchers of [63] used UAVs to ensure an ad hoc cooperation between vehicles. The collaboration provides reliable data delivery and ensures alternative routing paths in the case of link failures. Besides, authors of [64] proposed a connectivity recovery mechanism for UAV networks. UAVs can proactively find better positions to establish efficient cooperative communication links by leveraging on existing opportunities. Furthermore, researchers of [65] used UAVs as communication relays for data collection. They proposed a self-organized supervision and connectivity tool to perform automatic analysis and optimize the complexity of communication.

4.2. Using UAVs in cellular networks

UAVs can be used in cellular networks as (i) Aerial base stations, or (ii) Aerial user equipments.

4.2.1. UAVs as aerial User Equipments

UAVs can act as aerial User Equipments (UEs) within a cellular network in the coexistence of ground users. The cellular-connected UAVs can enable various applications such as real-time video streaming, area surveillance, item delivery, etc. Incorporating UEs UAVs in the cellular network generates several benefits. On the one hand, thanks to the worldwide accessibility of the cellular networks, it

is possible for the ground pilot to remotely control and command the UEs UAVs with unlimited operation range. Moreover, UE UAVs can be an efficient solution to keep connectivity between UAVs and the ground user equipments and the air traffic controllers. On the other hand, thanks to the advanced authentication mechanisms and cellular technologies, UEs UAV can achieve considerable performance improvement in terms of reliability, data throughput, security. Further, UEs UAVs are a cost-effective solution because they reuse the coexistent millions of cellular base stations without requiring to build new infrastructures.

However, the deployment of UE UAVs faces several practical issues such as flight altitude, mobility management and handover, interference management, cyber-physical attacks, energy efficiency, authentication, etc. To address these challenging problems, the research community is in full swing to develop mathematical and algorithmic solutions.

Authors of [66] analysed the impact of the implication of aerial UEs on the overall network efficiency. Simulations results show that by replacing a ground user by a UAV flying at 100 m, the throughput decreased to a factor 10, and the coverage drop from 76% to 30%. Therefore, to enable an adequate integration of UAVs into cellular networks, authors of [66] proposed an algorithm for optimal tilting of the directional antennas in UAV UEs allowing them to operate at higher altitudes. Simulations results show the increase of the coverage from 23% to 89%, and the throughput from 3.5 b/s/Hz to 5.8 b/s/Hz. The proposed mechanism is beneficial for sparse and medium dense cellular networks, and disadvantageous for very dense cellular networks.

Authors of [67] investigated an optimal inter-cell interference coordination technique and an air-ground performance trade-off. They maximized the weighted sum-rate of the existing ground users and the UAV by jointly optimizing the uplink cell associations and the power allocations over different resource blocks. They proposed a centralized inter-cell interference coordination technique to obtain locally an optimal solution, and they proposed also a decentralized scheme by dividing the cellular base stations into clusters to exchange data only between the cluster-head and the UAV.

Researchers of [68] addressed the security challenges for UAV-based multimedia streaming, UAV-based delivery systems, and UAV-based intelligent transportation systems. They introduced artificial neural network-based approaches. The proposed schemes allow the UAVs to use the wireless system resources with guaranteeing a secure operation. However, the learning algorithm generates large over-estimations that will result in a sub-optimal use of the resource allocation or a sub-optimal scheme of the UAV trajectory which will reduce the network performance.

4.2.2. UAVs as aerial base stations

UAVs as aerial base stations can provide cost-effective and reliable wireless communication for cellular networks. They can be supported by the connectivity of the existing terrestrial cellular networks. Compared to the terrestrial base stations, UAVs aerial base stations can adjust their altitude, enhance establishing line-of-sight communication links with the ground users, and avoid obstacles. Due to their flexibility, mobility, and adaptive altitude, UAVs aerial base station can efficiently complement the existing cellular systems by improving the network coverage to reach rural areas, and by giving the further capacity to hotspot areas.

However, the deployment of UAVs aerial base stations faces several practical issues such as placement and mobility optimization, power consumption optimization which have attracted the attention of the academic community. Authors of [69] proposed a physical layer security algorithm for multiple UAV-BSs providing wireless access to several ground UEs. They considered a group of eavesdroppers which are trying to attack the data delivery from the UAV-BSs to the ground UEs. The proposed scheme tries to maximize the secrecy rate by jointly optimizing the transmit beamforming and the UAV power consumption.

Authors of [70] addressed the challenge of the network throughput

and coverage improvement for deploying UAV-BSs. To solve those issues, they proposed a UAV-artificial bee colony mechanism for UAV-BSs deployment. The proposed algorithm can determine the optimal flight location of each UAV-BS to generate the maximum network throughput. Simulation results show that the proposed mechanism outperforms some bio-inspired algorithms in terms of the improvement of the network throughput and the enhancement of the UE coverage rate.

The UAV-BS trajectories are addressed in [71], in which a deep reinforcement learning algorithm is proposed based on the flow level model for learning the optimal traffic-aware UAV-BS trajectories. The proposed approach used only the starting UAV locations and the user traffic density to learn the optimal UAV-BS trajectories which maximize some cumulative performance metrics.

4.3. Using UAVs in satellite networks

Due to their global coverage, satellites are a viable option for UAV communications. It can relay data between separated UAVs and the ground gateways where the WiFi or cellular network coverages are unavailable. On the other hand, satellites can be exploited for the localization and navigation of the UAVs.

However, there are several limitations of using satellites for UAV communications. The most significant disadvantages are propagation loss and delay. Moreover, due to the power and weight constraints and the small size, UAVs can not carry the heavy energy-consuming communication equipment. Last but not least, satellite communication is characterized by the high operational cost which prevents their wide use for UAV communications. The aforementioned challenges should be addressed to perform satellite-enabled UAV communications. In this context, authors of [72] proposed the use of both the UAVs and satellites to ease the integration of the huge number of IoT solutions in the 5G framework. The proposed approach used satellites as relay nodes, UAVs as 5G-user equipments, and 5G-gNB has been located in the ground. The proposed approach permits to overcome the problems of the terrestrial infrastructure like the densification of the IoT devices and the restricted covered areas.

Researchers of [73] considered a multi-antenna multi-user system based on UAV and satellite cooperative network providing seamless connectivity to users. UAV is used as an aerial relay to transmit signals between satellite and several ground users based on amplify-and-forward protocol. The authors determined the maximal output SNR expression for the proposed architecture with an opportunistic user scheduling methodology. Based on the SNR expression, they derived expression of the outage probability for the proposed architecture. The outage probability expression in high SNR is exploited to determine the coding gain and the diversity order of the considered architecture.

Researchers of [74] addressed the issue of the resource allocation for an integrated UAV-satellite network. UAV-BS provides downlink connectivity to ground users who are not satisfied by terrestrial BS. The authors jointly considered resource management and the user association for the considered system. They assumed that the terrestrial backhaul links share resources with the satellite backhaul links and the terrestrial radio links. This joint is formulated as a competitive market setting where the data rate is optimized with a heavy ball iterative algorithm.

Authors of [75] proposed a secure UAV relaying algorithm for a hybrid terrestrial-satellite network in the presence of an eavesdropper. They considered three UAV Relay Selection (URS) methodologies based on the Closest(CURS), or Maximum(MURS), or Uniform(UURS) SNR. Then, they evaluated the impact of these strategies on the hybrid network in terms of the secrecy outage probability. Simulation results show that MURS secrecy performance is the best and UURS secrecy performance is the worst. Moreover, CURS and UURS methodologies do not enhance the secrecy diversity order.

5. UAV systems considerations

In this section, the most prominent UAV networks design considerations are discussed.

5.1. Mobility

UAVs networks are characterized by highly mobile nodes and diverse mobility models. The high mobility and speed of nodes cause changes in network topology and network partitioning. Nodes in UAVs networks are characterized by a high degree of mobility which results in several challenging communication design problems. The degree of mobility of UAVs is fully dependent on the application. For example, an earthquake struck, the UAVs would fly over the area of operation and must provide communication over links which would be slow or dynamic. Whereas, in applications, like agriculture or forest surveillance, the UAVs fly over a large area and the links are more dynamic.

Node density can be presented as the average number of nodes in a unit area. The distance between UAVs is generally large because UAVs are distributed in the sky. As a result of this, node density for UAVs network is small. Thus, the node density depends on the application scenarios and on the use case. The UAVs could move randomly or in an organized way (swarms) in two but or in three dimensions with rapid change in position. In some application nodes, mobility is not totally random and the nodes are probably visiting some known places. In these applications, the random mobility model, used in a large number of protocols, is not pragmatic. The probability estimation of UAVs visiting some places increases the delivering packet probability. UAVs network topology changes frequently because of the higher mobility degree of UAVs and the links outages. In fact, relative positions of UAVs may change and UAVs may wrongly work and be not covered by the network, so, the links may vanish. In the case of a UAV or a link failure, the other UAVs must complete the operations without disrupting user sessions. The major challenge is the selection of efficient neighbors and nodes arrangement to improve the performance of the networks.

5.2. Adaptability

Many UAVs network parameters can change during the missions. UAV network must adapt to changes and it should be capable enough to withstand the dynamic changes. The most prominent changes in such networks are the UAV locations, the distance between UAVs and the routes. Moreover, UAV failures are frequent in UAV network which generates the need for the UAVs injections. Besides, UAVs networks are affected by environmental conditions like the weather which disrupts of data links. In many applications, UAV system must adjust itself to any aforementioned changes or failures.

5.3. Reliability

The UAVs network while operating may be affected by various network attacks which can degrade the network performance. UAV network model should be fault-tolerant, robust and reliable. Thus, reliability depends on various parameters related to the mission task. The highly reliable network requires various delay-tolerant functionalities, recovery algorithms, and network pre-analysis. The reliability degree of UAV networks may vary according to the tasks. For example, an earthquake struck, higher reliability is required to the data transmission, whereas, the forward of pictures and video can be made with lower reliability for a certain applications.

5.4. Scalability

To reduce the cost of re-planning and re-organizing, the UAVs networks should allow adding new UAVs, therefore, it must be highly scalable. In fact, the scalability in the UAV network is related to the

nodes, and also to the resources. Performance and energy management are the most prominent parameters to evaluate the UAV networks. Recently, there are many researches that aim to extend the scalability of multi-UAV systems. In [17] authors proposed a solution to extend the range of UAV networks. The multi-hop communication in the UAVs link chain is used to extend the operation area. Besides, the obstacles in the operation area can influence the communication coverage of the infrastructure. The obstacle can be mountains, constructions, walls or buildings which may stop the radio signals between the ground base station and UAVs. Therefore, UAVs networks must extend their scalability by hiding the obstacles and by adding more relay nodes.

5.5. Latency

UAVs network must reduce latency and hind unpredicted delays. Latency constraint depends on the application. For example, in military monitoring, the mission over links must be done with minimizing delays. Also, to avoid collisions in multi UAVs networks it is required to decrease latency.

5.6. UAV platform constraints

Platform constraints are much greater in UAVs networks. There are two important constraints: the weight of the hardware and the space limitation especially for mini UAVs. The material weight influences on the network's performance. When the hardware is light, the payloads are lighter. Besides when the hardware is light, the number of sensors on the UAV and other peripherals can be increased. The space limitation, especially for mini UAVs, is very crucial for communication hardware which must be realized into the UAV platform [76].

5.7. Bandwidth and spectrum allocation

The bandwidth requirement depends on the application. Applications that require a high transfer rate must have high bandwidth. The selection of the appropriate network bandwidth is based on the tradeoff between energy-efficiency and bandwidth efficiency. For example, the applications using UAVs to collect data from distributed data sources in different areas and relay the collected data to a ground base station require high bandwidth. The application using advanced sensor technology can collect data with very high resolution, which requires higher bandwidth. Besides, the collaboration and coordination between UAVs require additional bandwidth resource, also, the efficient security in the application need much of the bandwidth. However, efficient use of the high bandwidth is constrained by the capacity of the communication channel, the velocity of UAVs, the error-prone structure of the wireless links and the miss of security with broadcast communication.

5.8. Power consumption and network lifetime optimization

UAV network power is based on the energy source of the UAV. However, small UAVs may typically have not enough power to fly and communicate during missions and they haven't the capability of refueling. It is required to design efficient configurations to control and decrease consumed power. Power consumption configuration influences the network lifetime because when the UAV power is drained, there are no transmitted signals and the links become intermittent. Especially, it is required to develop energy-efficient communication protocols to prolong the network lifetime. Equally, specific devices could be used for energy harvesting.

5.9. Localization

Localization accuracy is important when the UAVs network considers several UAV nodes. The cooperative network uses estimations

and approximations of the localization to perform complex tasks. GPS is the most known localization technology. However, UAVs are characterized by the high speed; therefore, it is required to accurate localization at the smaller time intervals. GPS is not sufficient for certain applications because it gives position information at the one-second interval. GPS must be equipped with an inertial Measurement Unit (IMU) to provide the position information at any time. Localization information must be exchanged over reliable and robust wireless links to ensure collision avoidance and the safety of devices. In [77] authors proposed a model using HAPS (High Altitude Platforms Station) for urban area monitoring. GPS is inefficient to exact localization information in the urban area. The authors used HAPS/UAVs as hot spots, which facilitate the use of TCP protocol to transmit localization data in urban areas.

5.10. Connectivity coverage

In [30] researchers considered two types of spatial coverage: area coverage and network coverage. In area coverage, UAV system provides some information from the environment such as surveillance, monitoring, and mapping. In this case, the UAVs are used to decrease the cost and the time and to ensure safety. Whereas in network coverage, UAVs are used as communication relays to provide connectivity between ground clients or collecting data from ground WSN nodes. The authors in [30] characterized several applications in area and network coverage domains based on the area sizes. There are many factors that influence coverage algorithms such as the mobility of the UAVs, the obstacles, the energy and the areas sizes.

In [24] the authors studied the network and the area coverage problems. The most important factor is the coverage ability which is the ratio of the coverage area. Some coverage algorithms can't cover the given area where the used number of the UAVs is inefficient to ensure the full coverage. Thus, it is required to develop an adequate UAV coverage solution depending on the UAVs motion. When the UAVs are used as static nodes, static coverage should be guaranteed. Whereas, when the UAVs are flying during the mission time, then, the dynamic coverage strategy should be considered. The area decomposition model and the path planning model must be estimated and developed to ensure efficient coverage. Thus, the mobility of UAVs can increase the coverage network ability; however, the high mobility of the UAVs makes the coverage algorithms very complicated. Moreover, the lack of energy in UAVs is the major issue of the coverage mission. It is required to design efficient configurations to prolong the lifetime of the UAVs network and to reduce the coverage holes. The terrain over which the UAV is flying contains obstacles such as buildings, trees, mountains, so, the UAVs must have the capability to detect an obstacle, or it must consider a safe path to avoid collisions with objects scattered in the flight terrain. The variation of the altitude of flight can be a solution to avoid obstacles but the variations of the UAVs flight altitude lead to several problems [78].

5.11. Security and privacy

UAV networks are considered as a Cyber-Physical System (CPS). The physical domain interacts with the cyber domain using sensors, actuators, communication equipment, etc. The cyber domain represents the UAV closed loop which is composed mainly of four components: (i) sensing, (ii) communication, (iii) computation, and (iv) control. The variety of components lets UAV networks vulnerable to many external attacks because security problems in any component will paralyze the system.

- **Sensing:**
UAVs are equipped with several remote sensing devices to introduce the data from the physical world into the UAV network. Sensor spoofing attacks are steered towards on-board sensors such as the

GPS receivers, vision, lidar, radar, sonar, and IR sensors. An adversary can blind the vision sensors, or send fault-information through the GPS channels. As the UAV autopilot is relied on sensor information for guidance and navigation, the corrupted sensor can be very dangerous. Sensor spoofing attacks can cause mode confusion and overriding gains through hacking.

- **Communication:**
Communication permits the data to circulate inside of the UAV network. It guarantees the data distribution and sharing. The main features that make the UAVs networks vulnerable to attack are presented in the following. Firstly, the wireless links can cause links attacks: passive eavesdropping, message misuse, active interfering, leaking of secret information, message replay, data tampering, and denial of service. Secondly, the uncontrolled environment can make link attacks from the outside and even the inside. If an attacker is in the UAV transmission range it can transmit and receive packets. Thirdly, the dynamic topology makes the detection of the misbehaving node difficult. Moreover, during the route discovery process, UAVs broadcast the packets without taking attention to the receiver. No guarantee that the generated path is free of attack nodes.
- **Computation:**
Computation is exploited for analyzing and decision making using all the acquired data. UAVs have limited storage and computing capacities. An attacker can launch the sleep deprivation attack that tries to drain UAV battery. Moreover, an attacker can produce a flooding attack to exhaust the network bandwidth and consume UAVs and GCS resources.
- **Control:**
Control translates the decisions to instructions for the actuators of the physical world. An attacker tries to take control of the UAV, it can act as a UAV flight controller to enable the injection of the desired values.

Privacy issues occur with security problems. Due to their ability to video capture in many angles and areas, UAV can cause a very crucial issue. Such information can be exploited to profile any individual. Therefore, it is required to develop new on-board techniques to guarantee privacy while capturing videos.

6. UAVs related protocols and mechanisms

6.1. Routing protocols

In the combined literature, there are several routing protocols that are designed for MANET. UAVs networks are a sub-class of MANET networks, and it has specific characteristics such as the high mobility of nodes, the frequent changes in the quality of links and the energy constraints. Therefore, several MANET routing protocols are adapted to the UAVs networks, and there are, also, some specific routing protocols proposed specifically to the UAVs networks. These protocols can be divided mainly into three categories: 1) Topology-based routing protocols 2) Position-based routing protocols 3) Swarm-based routing protocols. Fig. 7 illustrates this classification.

6.1.1. Topology-based routing protocols

Topology-based routing protocols can be divided mainly into four categories: (i) static, (ii) proactive, (iii) reactive, and (iv) hybrid.

A. Static routing protocols

In static routing protocol, a routing table is computed and uploaded to UAV nodes when the operation starts and cannot be modified during the mission. Every node can communicate with some UAVs.

a. Data-Centric Routing (DCR)

The routing protocol DCR [16] can be used in UAVs networks. The routing algorithm is effectuated with respect to the data

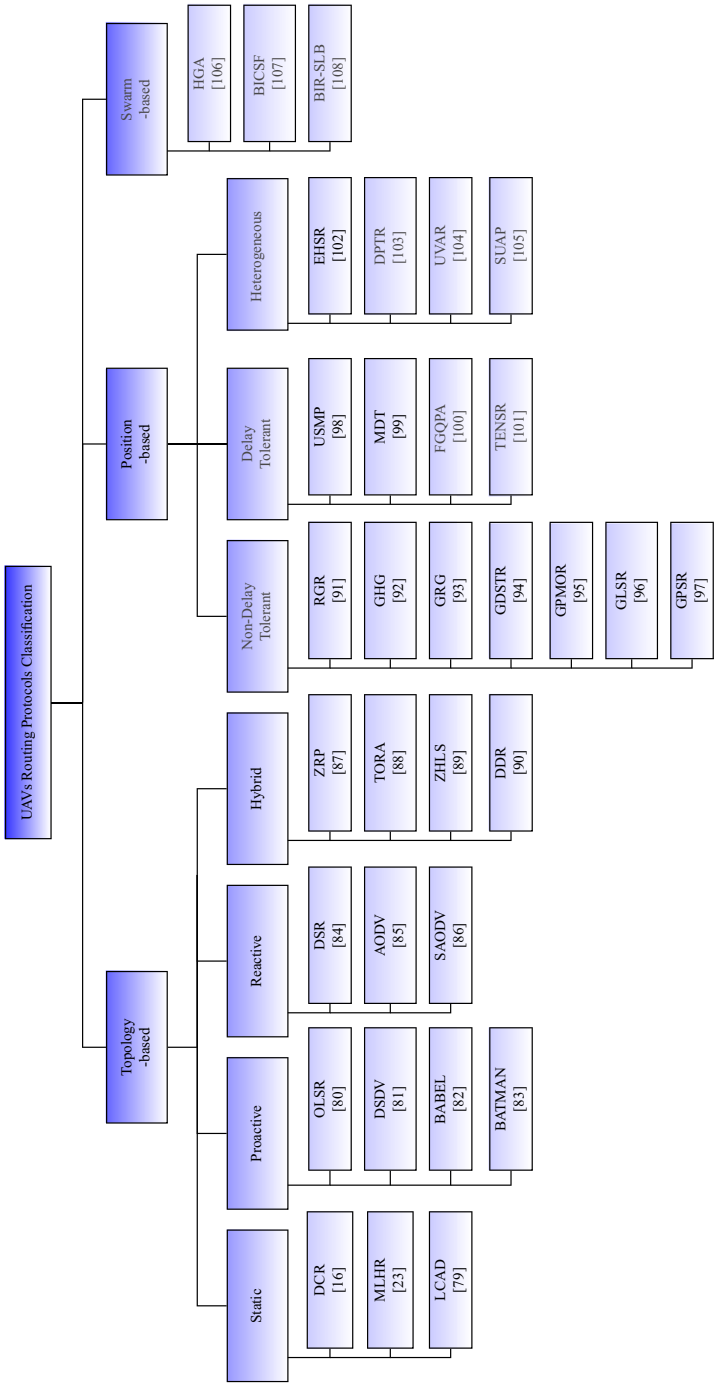


Fig. 7. Routing protocols classification .

content, it ensures the one-to-one and the one-to-many data sending [79]. When the data is needed by a set of nodes, this routing strategy is used. The data generator, which is called the publisher, is connected automatically to the data user, which is called the subscriber. The producer starts the data dissemination and it selects the data which must be published. Data aggregation algorithms are used in Data-Centric broadcasting for energy conservation. This routing strategy performs three scopes of decoupling: (i) Space decoupling, (ii) Time decoupling, and (iii) Flow decoupling.

b. Multi-Level Hierarchical Routing (MLHR)

In MLHR [23] UAV networks are organized hierarchically by clusters operating in large mission areas. The cluster head diffuses data to the inter-cluster nodes which are in its communication range. The cluster head can be changed but it will impose a large overhead on the UAV networks. This routing model is used in the networks where the UAVs are controlled in different swarms.

c. Load Carry and Deliver Routing (LCAD)

LCAD [80] is one of the earlier proposed protocols for UAVs networks. In this mechanism, the ground node transmits data, which can be the path planning, to the UAV, then, the UAV flies and transfers the data to the destination ground node. The main purpose of LCAD is to improve the security and maximize throughput, but, using a single UAV increase the delays. Therefore, to solve this issue, the multi UAVs systems are deployed considering multi-sources and multi-destination. LCAD is used in bulk data transfer applications and delay-tolerant applications.

B. Proactive routing protocols (PRP)

PRP is based on routing deployed to collect all routing information of a specific region in the network. The routing tables must be updated when the network topology change. The major advantage of proactive routing protocols is that it has the recent information of the routes; so, it is not difficult to select routes from the transmitter to the receiver. Whereas, PRP suffers from some explicit disadvantages. Firstly, there is the need of exchanging some messages between nodes, therefore, the bandwidth utilization cannot be optimized, and these types of protocols are not efficient in highly mobile networks wide networks. Secondly, UAVs networks are characterized by highly changing in the topology and frequent occurrences of failures and these types of protocols have a slow reaction.

a. Optimized Link State Routing Protocol (OLSR)

In OLSR [81] two types of messages are considered which are hello messages and topology control messages for neighbors discovering. Hello messages are exploited to detect the neighbor nodes which are in the direct communication range. This message is broadcast periodically to one-hop neighbors. The topology control messages are exploited to update system topological information. At startup, OLSR determines routes to all destinations and it maintains them periodically by a refresh process. This periodic flooding mechanism causes large overheads. To solve this problem, Multi-Point Relay (MPR) algorithm must be used. Each node chooses its own MPRs neighbors and it receives the routing messages only from those nodes. Link-Quality OLSR (LQ-OLSR) protocol [82] takes into account the wireless links quality using the ETX (Expected Transmission Count) metric. The ETX metric measures the expected number of transmissions that must be to send a packet successfully over a link. These changes do not require modifications in the format of the control message.

To improve the selection of routes, Directional OLSR [83] is proposed. This protocol aims to reduce MPR numbers. MPR selection mechanism is based on the global profile which contains information collected from the first three layers. With directional antennas, DOLSR reduced the number of MPRs [84]. Besides, researchers in [82] proposed Predictive-OLSR routing protocol

which aims to find efficient routes in very dynamic UAV networks by using GPS information. The predictive-OLSR uses the ETX metric, taking into account the relative speed between the nodes. Simulation results show that the Predictive-OLSR outperforms OLSR providing reliable communication in very dynamic conditions. In [85] Mobility and Load OLSR (ML-OLSR) protocol are proposed. ML-OLSR protocol introduces the mobility and the load-aware algorithms, therefore, the control packet formats are improved. Simulation results show that ML-OLSR performs better than the OLSR protocol with regards to the average end-to-end delay and packet delivery ratio.

b. Destination-Sequenced Distance Vector (DSDV)

DSDV [86] uses usually Bellman-Ford algorithm with few modifications to be more appropriate for UAVs networks. Each node considers a routing table that has a sequence number for all other nodes. When the topology of the network changes these changes are diffused via the update mechanism, so, it requires large network bandwidth for procedures of updates. DSDV is unsuitable for dynamic networks and it does not support the multipath routing. Moreover, DSDV needs high resources for computing and storage. In [87] researchers compared the performance of DSDV with other protocols. Simulation results demonstrate that the DSDV packet delivery ratio is low especially in higher mobility environments. To resolve this problem, authors in [88] proposed a novel message exchange mechanism to generate invalid route reconstruction. This mechanism disseminates invalid route information in such areas when a link fails, without affecting the total network. The procedure of invalid route reconstruction must ensure that the novel route is loop-free. In [89] researchers compared the performance of I-DSDV with DSDV and another protocol. Simulation results show that I-DSDV reduces the dropped data packets compared to DSDV, with little increase in the overhead at higher node mobility.

c. **BABELBABEL** [90] is used in unstable networks because it limits the frequency and events duration such as black holes and routing loops during re-convergence. These provisions are based on the link quality estimation. Even, when it detects mobility, it quickly passes to a loop-free configuration. One of its problems is that it generates more traffic than other protocols when there are changes in the network topology. In [82] authors demonstrated that BABEL is unsuitable in the UAV environment based on the average outage time and the datagram loss rate.

d. Better Approach to Mobile AdHoc Network (BATMAN)

BATMAN [91] maintains proactively information about the associated nodes to the network, nodes which are identified are accessible by single-hop or multi-hop communication links. BATMAN does not calculate the total route, for each destination it cares about only the best next-hop. It is efficient and very fast in terms of implementation because its packets are very small and contain only a limited amount of information. Besides, communication links quality varies in terms of data rates, packet loss, and interference. Nodes continuously disseminate origin messages (OGMs); and when node information is well known, this node is used in routing decisions. As BATMAN depends on packet loss it is unsuitable for reliable wired networks. There is another problem which is the knowledge decentralization about the best route in the network, each node doesn't have all the data. To resolve this problem, a collective intelligence network is created. In [92] researchers demonstrated that BATMAN-adv is loop-free and reliable. In [93] simulation results illustrate that BATMAN-adv performs well compared to BATMAN.

C. Reactive routing protocols (RRP)

In RRP, a route between two nodes is stored when there is the transmission of packets between them. RRP is considered to reduce overhead generated by proactive routing protocols. In this routing protocol category, there is no periodic exchange of messages which

make the bandwidth efficient. Whereas, the process of finding routes may take a long time which causes high latency. RRP can be categorized into two types: hop-by-hop routing and source routing. In source routing, for each data packet, the complete route from source to destination address is transported, therefore, this information is used by intermediate nodes to forward packets. In hop-by-hop routing, every data packet transports only the destination and the next-hop addresses. The routing table to forwarding data is maintained by intermediate nodes. The advantage of this method is that routes are flexible to the frequent environment changes. The disadvantage of this method is that every intermediate node may store and keep routing information for every active route; moreover, every node must be aware of their neighbors. In this routing model, there are two types of messages: Route Request and Route Reply messages. Route Request messages are created and disseminated by the source node, Route Reply messages are the responses from the destination node to the Route Request messages.

a. Dynamic Source Routing (DSR)

DSR [94] is created mainly to multi-hop wireless mobile mesh ad hoc networks. When the source nodes have data to be sent, it disseminates route request messages to their neighbor nodes that are in their transmission range. Therefore, there are many Route Request messages, to avoid confusion, the source node must add unique request-id. DSR designs the entire hop-by-hop route to the destination node, if the source node can't use the current route, because of the modifications in the network topology, then the route repairs and route discovery algorithms are activated. In [95] DSR protocol was implemented to find a new route in the UAV network.

b. Ad-hoc On-demand Distance Vector (AODV)

AODV [96] has similar characteristics with DSR. The only difference is in the implementation of the routing table. When using DSR each node stores several entries in the routing table for each destination while in AODV; for every destination, there is only one record. Moreover, in DSR all the data packets have to transfer the complete route from the source node to the destination node. Whereas, in AODV, the source node saves only the next-hop data communication. Therefore, routing packets in the network are decreased. AODV protocol is generally composed of three phases: the discovery route, the packet transmission, and the route maintenance. Besides, AODV considers three message types: Route Request, Route Reply, and Route Error. AODV is studied for the adaptation to the UAV networks. During the route construction, there are many delays. Especially, during route discovery, link or node malfunction may trigger that introduces additional delays and consumes further bandwidth. Besides, intermittent links are more pervasive which decreases the throughput dramatically.

c. Time-Slotted On-demand Distance Vector (SAODV)

Time-slotted on-demand routing protocol [97] is a time-slotted version of AODV. In this protocol, each node has its time slot in which it can send the data packet. This protocol reduces the use of network bandwidth, whereas, it increases the packet collisions and the packet delivery ratio.

D. Hybrid Routing Protocols (HRP)

HRP is essentially a combination of the aforementioned protocols, and it presents a compromise between the huge overhead of PRP and the high latency of RRP. HRP is applicable in large networks by dividing the entire network into a number of regions wherein the intra-zone PRP is used and in the inner-zone RRP are used.

a. Zone Routing Protocol (ZRP)

ZRP [98] is based on the zones concept according to distances between nodes. If the source and the destination nodes belong to the same zone, then, the source node can transmit data instantly. When it is required to send data packets outside the zone the reactive method is applied to determine a route to the required

destination. This protocol decreased significantly the amount of overhead compared to proactive protocols. It also reduced the associated delays compared to pure reactive protocols. Whereas, dividing the routing zone into large values let this protocol behave like a proactive protocol, and, for small values, this protocol behaves like a reactive protocol. Therefore, the size of the zone should be selected carefully because it impacts the performance of the routing protocol.

b. Temporarily Ordered Routing Algorithm (TORA)

TORA [99] is used in multi-hop networks, its routers preserve only information about adjacent. It tries to decrease the control message propagation in the highly dynamic environment, by reducing the reactions to topological changes. It builds and keeps a Directed Acyclic Graph (DAG) starting from the source node arriving at the destination node. DAG generates multiple paths between these nodes. When there are broken links, it is easy to find new routes which increase the adaptability. TORA does not employ the shortest path solution, it uses frequently longer routes to reduce network overhead.

c. Zone-based Hierarchical Link State (ZHLS)

ZHLS [100] employs hierarchical structure and it divides the network into non-overlapping zones, every node has a zone ID and a node ID. The hierarchical topology is composed of two levels: zone level topology and node-level topology. Similarly, ZHLS has two categories of link-state updates; one is the zone level LSP (Link State Packet) and another is the node level LSP. Periodically, each node broadcasts the node-level which contains the neighbors IDs in the same zone and all other zones IDs, to all nodes in the same zone. Consequently, all nodes in the same zone keep the same node level link state data. When there is a new or a broken virtual link, the gateway nodes disseminate the zone level LSP throughout the network. In ZHLS, there is no cluster-head or location manager selection, so, there are no associated overheads and traffic collisions can be avoided. In this protocol, when it is required to find a route to a destination, the source node disseminates a zone-level location demand to all other zones. Thus, ZHLS reduced communication overheads compared to the reactive routing protocols which generate flooding packets. Moreover, in ZHLS the routes are adaptable to the frequent topology changes because only the zone ID and the node ID of the destination are required for routing. However, in ZHLS each node must have a static zone map, which may not be realizable in applications having a dynamic network geographical boundaries.

d. Distributed Dynamic Routing (DDR)

DDR [101] is a hierarchical routing protocol. In DDR the hierarchy is constructed using periodic control messages that are exchanged between neighboring nodes only. The trees in the network are connected together by the gateway nodes. Each tree forms a zone having a zone ID which is generated via a zone naming algorithm. To determine a stable route between the sender and the receiver, DDR uses the inter-zone and the intra-zone routing tables. DDR does not rely on only static zone map and it does not necessitate a cluster head to transmit data and control packet between different zones and nodes. Whereas, the channel contention may increase at the preferred neighbor which causes much greater delays. Moreover, networks having high traffic, DDR may reduce the throughput, because many packets will be dropped when buffers are full.

6.1.2. Position/geographic-based routing protocols

Geographic-based routing protocols require physical position information of the active nodes in the network. Usually, each node provides its own location by using GPS or another type of location mechanism. Position-based routing protocols can be divided mainly into three categories: (i) non-delay tolerant routing protocols, (ii) delay tolerant routing protocols, (iii) heterogeneous routing Protocols.

A. Non-Delay Tolerant Routing Protocols

a. Reactive Greedy Reactive (RGR)

In [102] RGR is proposed for applications containing many UAVs interacting between themselves without static infrastructure. UAVs consider UAV-to-UAV communications to reduce mission delay and improve reliability in aerial networks. It combines GGF and the reactive routing protocol to be appropriate to restrict topological characteristics of UAVs networks e.g. the high mobility. This protocol uses location information of UAVs like the reactive end-to-end route in the routing process. In RGR, when the source node doesn't find a legal route to transfer data packets, it triggers a route discovery mechanism, by disseminating Route Request (RREQ) packets in the network. When the source node gets the Route Response (RREP) packet from the destination node, data packets saved in the source node are transmitted. In RGR, the destination position information is attained by each intermediary node in the RREP packet. In the route maintenance algorithm, if an intermediary node doesn't obtain three successive HELLO messages, the link is supposed lost and the reactive route is assumed breaded. Then, RGR avoids the reactive mode and employs the GGF mode. Simulation results demonstrate that RGR outperforms the other protocols such as AODV in terms of packet delivery ratio and delay.

b. Greedy-Hull-Greedy (GHG)

GHG [103] dissociates the entire network space to several closed sub-spaces to perimeter the local recovery process. GHG is based on the network hulls which are created by Partial Unit Delaunay Triangulation algorithm (PUdT). Like planarization, the Delaunay Triangulation distributed computation is a hard problem, therefore, GHG is not suitable for networks having arbitrary topologies.

c. Greedy-Random-Greedy (GRG)

In GRG [104] messages are disseminated greedily until a local minimum is encountered. The virtual cubes technique is used to locally capture the space of holes in the 3D network. To search in a virtual cube, the authors used and analyzed many randomized recovery algorithms like a region-limited random walk, random walk on the graph, random walk on the surface, bounded flooding and bounded DFS on spanning tree. The simulation result shows that DFS on spanning-tree performs well other algorithms in sparse networks and random walk algorithms outperform for denser networks.

d. Greedy Distributed Spanning Tree Routing-3D (GDSTR-3D)

GDSTR [105] used a 2-hops communication range to find the closest neighbor during greedy forwarding to overcome the probability of local minima. If local minima still continue in a 2-hop communication range, the hull tree routing mode is used to discover a node having the destination in its child node. GDSTR finds the shorter routes and creates less maintenance traffic. It also offers deterministic routing and simulation and tested experiments show that this protocol guarantees packet delivery and generates significantly lower overhead. Whereas, in this protocol, each node must keep a set of hulls which increases the storage space that is proportional to network size. Consequently, some nodes can be heavily loaded.

e. Geographic Position Mobility Oriented Routing (GPMOR)

Traditional position-based protocols use only the node's location information. But, GPMOR [106] predicts the UAVs movement with the Gaussian-Markov mobility pattern, this information is used to find the next hop. GPMOR maintains fewer hops whereas; GLSR and GPSR select the next-hop just by the distance. GPMOR provides efficient data forwarding in terms of packet delivery ratio and latency compared to the position-based routing protocols such as GPSR and GLSR. The packet delivery ratio of GPSR and GLSR degrades immediately when the velocity increased because those protocols do not consider the impact of the high

mobility that causes poor connectivity.

f. **Geographic Load Share Routing (GLSR)** GLSR [107] uses a geographic load share routing metric. It exploits the path variety to compensate congestion issues based on the TDMA technique. GLSR maximizes the speed of all neighbors to the destination. GLSR maximizes also the use of the existing capacity for the Air to Ground (A2G) link by exploiting the position of the UAV and its buffer size. But, links in the UAVs network are usually very short because of the high mobility of the UAVs, which decreases the service quality. Besides, GLSR supposes the topology of the UAVs network is static that limits its application.

g. Greedy Perimeter Stateless Routing (GPSR)

GPSR [108] is a position based routing protocol; this protocol uses the closest geographically neighboring nodes to the destination to forward packets. Every GPSR node keeps a planar graph of all nodes which are in its transmission range. Then, the full network graph is explored; the vertexes of this graph are all the nodes in the network and the represented edges between each couple of nodes mean that they are capable to communicate. To ensure that nodes have the neighbor information, every transmitted packet includes the location information. Periodically in the absence of data packets, beacon packets containing location information are broadcasted. In [109] authors created a simulation framework to analyze the performance of geographic-based routing protocols and their outcome that GPSR is suitable to compactly positioned UAVs network. But, reliability is the main issue of this protocol.

B. Delay Tolerant Routing Protocols

a. UAV Search Mission Protocol (USMP)

USMP [110] is a protocol for UAVs searching in a 2-dimensional grid. It combines inter-UAV communication and geographic routing. USMP aims to improve search in terms of distance traveled by UAVs, the decrease of UAV direction changes and the total searches. USMP location update feature considers two design techniques. In the first design, there are messages which are explicitly transmitted to neighbor nodes, and in the second design, GPSR location information is used. The path selection is performed by using the reservation message. The implementation of USMP is based on several parameters such as swarm size, transmission power, initial location and sensor type. The authors studied the performance of USMP and GPSR protocols and simulation shows that GPSR degrades performance in total searches and distance traveled, but it improves direction changes. Whereas, USMP improves performance compared to scenarios which do not contain inter-UAV communications.

b. Multi-hop Delaunay Triangulation (MDT)

MDT [111] provided guarantee delivery to each connected network of nodes in 3 dimensions, and it ensures efficient forwarding of packets from local minima with low routing stretch. The authors analyzed the performance of their protocol and simulation results show that MDT generates better routing stretch when compared to a number of protocols including GRG and GDSTR-3D. Moreover, the authors show the suitability of MDT for dynamic topologies with variation in the number of links and nodes.

c. **Fountain-code based Greedy Queue and Position Assisted (FGQPA)** The objective of FGQPA [112] is to reduce the delay of transmission within the UAV network. It considers two modules which are Power Allocation and Routing (PAR) module and nearest span module. PAR aims to transmit the data packets based on UAVs having a low queue backlog. The second module aims to transmit the packets to the destination when the transmission delay is shorter. Thus, each UAV has a local queue and it defines its own position which is exploited as a routing metric.

d. **Tactical Edge Network Social Routing (TENSUR)** In TENSUR [113], the UAV mobility plans are estimated at each time. This

Table 5
A comparative study of UAVs routing protocols.

	Basic Concepts	Suitability to UAVs	- Advantages	Limitations
Topology-based	Static	-	Two kinds data sending Energy conservation	Based on data content No support of high mobility
		-	Support several UAVs Maximize throughput	Large overhead No support of high mobility
	Proactive	-	Improve the security Maximize throughput	Increase the delays No support of link breakages
		-	Having recent routes information	Increase overheads Increase bandwidth utilization
		-	Having recent routes information Remove routing loops	Require large network bandwidth Low packet delivery ratio
		-	Does not calculate the total route Small transmitted packets	High packet loss ratio Frequent variation in the communication links
	Reactive	-	Used in multihop ad-hoc networks Possibility to change the route	Large memory space Large bandwidth utilization
		-	Each destination consider one route Saves only the next-hop the data communication	Route discovery introduces additional delays and consumes much bandwidth
	Hybrid	-	Decrease the congestion an bandwidth	Introduce complex computations
		-	Decrease overheads Reduce delays Enhance the nodes organization	The constraint of the zone size inadequate for high mobility
Position-based	Non-Delay Tolerant	+	Decrease control packets propagation Reduce overhead	No use of the shortest path solution Large memory space dues to the multi-paths exploration
		+	Minimize traffic collisions Reduce communication overheads	Each UAVs must have a static zone Increase control packets propagation
		+	Select relays using predictions	No consideration of fragmentations
		+	Uses the closest neighboring Frequent broadcasting of location information	Low reliability No support of link failure
		+	Increase packet delivery rate Decrease latency	Maintains fewer hops No consideration of stability of links
	Delay Tolerant	+	Increase the delivery ratio Decrease the traveled distance	Increase the delivery delays
		+	Reduce delivery delay and packet losses	No recovery process
		+	Enhance delivery ratio and delay	Specific use case Energy overuse
		+	Enhance throughput and delay	No consideration of ground mobility
	Heterogeneous	+	Fully exploits UAVs	Cause high delay Energy overuse
		+	Avoid malicious modifications Maintain links with the neighboring UAVs	No deal with high the mobility
		+	Enhance memory storage Low computing time	Complex behavior modelling
		+	Low energy consumption Enhance cluster lifetime	Decrease throughput
		+	Find the best path Decrease control overheads	Consume energy and bandwidth
		+		
		+		
	Swarm-based	+		
		+		
		+		
		+		
		+		

information is transmitted to all UAVs via a dedicated channel. The probability of having two UAVs in range of each other is studied in a graph compound of vertices (UAVs) and weighted edges to define the tie strengths between each pair of UAVs. The routing process composes two main parts which are the exchange of routing table and route selection. The goal of exchanging the routing table is to provide all UAVs the required information to estimate the adjacency probabilities which are used to select the appropriate route.

C. Heterogeneous Routing Protocols

a. Extended Hierarchical State Routing (EHSR)

EHSR [114] is used in hierarchical heterogeneous multi-layer ad hoc wireless networks. It establishes multi-level communications with multi-interfaces at different levels. The EHSR is the extension of the HSR and it solves many problems caused by the multi-hopping. Simulation results show that EHSR improves the scalability by decreasing the number of transmissions through the hierarchical multi-layer infrastructure help.

b. Distributed Priority Tree-Based Routing Protocol (DPTR)

DPTR [115] solves the network sparseness by using the features of the Red-Black (R-B) trees. DPTR provides different control packets over different channels and rates. It starts by identifying aerial and ground nodes and interfacing based on neural structure. Then, it considers the R-B tree to make communication more efficient. Finally, three cooperative frameworks are designed: (i) Ground Ad hoc Network, (ii) Aerial Ad hoc Network, and (iii) Cooperative Network Framework.

c. UAV-Assisted VANET Routing Protocol (UVAR)

UVAR [116] uses the existed UAVs in the sky for the connectivity estimation degree and data delivery. For each segment, hovering UAVs assign a score based on four metrics: (i) the connectivity between the two intersections, (ii) the traffic density, (iii) the distance separating the communicating vehicles, and (iv) the vehicles real distribution. This score permits us to avoid the different obstacles and to support the routing process.

d. Secure UAV Ad-Hoc Routing Protocol (SUAP)

SUAP [117] adopts geographical leases to calculate the correlation between the packets transited distance and the hop count. This is done by keeping the links with the neighbors. SUAP makes some verifications to detect any wormhole links based on the computed correlations.

6.1.3. Swarm-based routing protocols

The term Swarm Intelligence means optimizing and solving problems based on nature-inspired behavior. It provides optimal solutions for UAV routing.

a. Hybrid Genetic Algorithm (HGA)

HGA [118] supports the cooperation of multiple UAVs and a ground vehicle for efficient parcel delivery. The routing algorithm allows the UAVs to simultaneously deliver many parcels to multiples destinations residing in different locations. HGA consists of a pipeline of many modules: heuristic population initialization, population management, and population education. The evaluation results show that HGA has significant effectiveness over existing algorithms.

b. Bio-Inspired Clustering Scheme for FANETs (BICSF)

BICSF [119] uses the hybrid mechanism composed of Glowworm Swarm Optimization (GSO) and Krill Herd (KH). Firstly, a cluster management algorithm is performed based on KH behavior. Then, the UAVs start transmitting the data to the cluster head which is responsible for sending the data to the destination. For efficient routes election, the path detection function is performed based on the number of neighbors, the weighted residual energy, and the distance inter-UAVs.

c. Bio-Inspired Routing with Shortest path and Load-Balancing

(BIR-SLB)

BIR-SLB [120] inherits the AntNet routing behavior with some metrics extensions for the multi-objective optimization. To find the optimal route two kinds of packets are considered: (i) Forward Exploration Ant (FEA) and (ii) Backward Information Ant (BIA). FEA packets are used to explore quickly the network. BIA packet is used to find the best path based on the defined metrics.

6.1.4. A comparative study of UAVs routing protocols

Based on the different studied UAVs routing protocols presented in the past section in this paper, we have identified the advantages, the suitability, and the limitations of each routing protocol. The results of these investigations are regrouped and highlighted in Table 5.

6.2. Access protocols

The high mobility of nodes, link quality changes and packet latency are the major design problems in MAC layer design for UAVs network. Several UAVs MAC protocols have been proposed by researches, accordingly, we have elaborated two classifications. The first classification is based on the antenna where we distinguish between approaches using an omnidirectional antenna, approaches using a directional antenna, and hybrid approaches. The second classification is oriented access mechanisms; we classified the related mechanisms into five categories: 1) Code Division Multiple Access (CDMA)-based protocols, 2) Time-division multiple access (TDMA)-based protocols, 3) Non-Orthogonal Multiple Access (NOMA)-based protocols, 4) hybrid protocols, and 5) cross-layer protocols. Fig. 8 illustrates the two classifications.

6.2.1. Classification based on antenna categories

A. Omni-directional antenna mechanisms

Antennas are used to radiate or receive energy between all directions. Omni-directional antennas are used in multi-hop wireless networks because of their low cost and their limited size. Whereas, omnidirectional antenna mechanisms have many disadvantages. The most important limitations are the low gain of an antenna and the limited communication range. The authors in [121] proposed a collision-free MAC (CF-MAC) protocol which allows the UAVs using omnidirectional antennas and half-duplex radio. To reduce the collision probability they utilize a region marking scheme which can reduce the hidden terminal issue without any RTS/CTS. In [122] the authors proposed two broadside omnidirectional antennas. These antennas have smart hardware which is easy in the installation. They generate a significant gain in the network which is very useful in enhancing the Line Of Sight (LOS) range of UAVs. Researchers

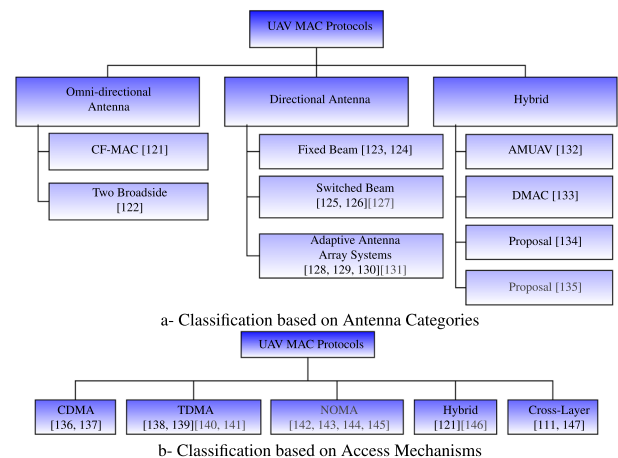


Fig. 8. UAVs MAC protocols classification.

compared CF-MAC with VeMAC protocol which is a MAC protocol for VANETs. Simulation results show that CF-MAC is more efficient and it can improve the channel access, and it reduces the collision probability.

B. Directional antenna

a. Fixed beam

In [123] authors used a fixed beam antenna arrays installed in UAV wing. They demonstrated that the antenna array generates the quality of the signal, measured on Received Signal Strength Indicator (RSSI), better than the single wire antenna, which extends the flying communication range. Besides, authors in [124] used a fixed beam directional antennas arrays in UAVs. They studied the performance of this system using software simulations. Researchers used the UAV rotation to point a beam to the desired direction, which improves the system capacity when compared with the omnidirectional transmission. However, this approach has some disadvantages including the rotation time of the UAV and the need to modify the MAC protocol.

b. Switched beam

In [125] authors incorporated the use of neighboring nodes location estimation and the directional antennas in the MAC layer. Thus, they use UAVs equipped with a pair of transceivers and every transceiver uses switched beam directional antennas. LODMAC improves spatial reuse and network capacity. This protocol added a new packet Busy to Send (BTS) to resolute the directional deafness problem. Authors compared their protocol with DMAC protocol [126] simulation results show that LODMAC outperforms DMAC protocol in terms of throughput and average network delay and it suitable for UAVs network. Moreover, in [127] the authors presented implementation and experimental evaluation of switched-beam data link antenna in the UAV. Moreover, in [128] the authors proposed a Backhaul Link with Antenna Steering for UAV system. The UAV is carrying a microwave antenna and a switched beam mechanism to keep the antenna steered toward the ground station. Experimental results demonstrate of the proposed scheme.

c. Adaptive Antenna Array Systems

In [129] the authors designed and implemented a multibeam adaptive array antenna system for the UAVs network using Field Programmable Gate Arrays (FPGAs). The authors used Fast Fourier Transform (FFT) to quantify the increase of the required FPGA resources and the computational load. Moreover, the authors provided an example of the Multibeam Smart Antenna design based on the Xilinx Virtex-7 FPGA. This example is designed with the consideration of the implementation complexity, the system performance gain, and the required hardware. In [130] the authors designed two beamforming and scheduling scheme for UAVs networks. They considered a UAV central node (UAV-C) having multi-antenna to collect data from many other distributed UAV nodes (UAV-D) which have single-antenna. The first scheme is considered when the channels between UAV-D and UAV-C are both perfectly known; in this case, the authors used the Minimum Mean-Square-Error (MMSE) beamforming in combination with a combinatorial approach. Whereas, the second scheme is considered when the channels between UAV-D and UAV-C are subject to the estimation errors; in this case, the authors proposed to use robust MMSE beamformers in a combinatorial manner that make the beamformer robust to the channel estimation errors. Simulation results show that the proposed schemes improved the gain in average throughput compared to non-robust and robust beamformers without scheduling and non-robust beamformers with scheduling.

In [131] the authors developed a framework for attitude estimation using 3D Line-Of-Sight (LOS) vector which is obtained from the antenna array. They considered a base station sending a signal to a UAV, the signal is diffused with a finite speed to every

receiving antenna, and every antenna receives the same signal with a differently delayed version. Therefore, the inter-antenna spacing and the signal propagation speed are known. Then, it is easy to find Direction-Of-Arrival. Moreover, in [132] researchers proposed a wide beam coverage antenna array for UAV to everything (U2X) communication. It is composed of two dipole antennas arrays with parasitic elements. Each dipole antenna contains a broadside beam. The parasitic elements ameliorate the beam coverage. The proposed antenna has broadened beamwidth and high gain. It has a miniaturized size and symmetrical structure. Therefore, it is easy to mount it to the UAV.

C. Hybrid

In [133] the authors proposed an adaptive MAC protocol AMUAV for UAV networks using directional and omnidirectional antennas. Every UAV is equipped with four antennas: two omnidirectional and two-directional. One of the directional antenna is marked primary and it is placed above the UAV and the second one is marked secondary and it is placed under the UAV. Before starting the sending of the data packet, the AMUAV scheme measures the distance between the UAVs. When the distance is lower than the omnidirectional antenna range, then the data will be transmitted using an omnidirectional antenna, else the AMUAV check the UAVs altitude to choose of the directional antenna the primary or the secondary antenna). Therefore, AMUAV adapted the directional antenna and external parameters attached to the UAV. The authors compared their scheme with the IEEE802.11 protocol. Simulation results showed that AMUAV improves the network performance.

In [134] the authors proposed a hybrid antenna scheme (D-MAC), D-MAC used the directional antenna for transmitting the DRTS packets to a particular direction, and OCTS packets are diffused using an omnidirectional antenna. After the successful transmission of the DRTS and OCTS packets, the sender sends the data packet using a directional antenna, after the successful reception, the receiver node sends an ACK using the directional antenna.

In [135] the authors proposed a hybrid routing scheme for the UAV network which combines the omnidirectional and directional transmissions utilizing location and trajectory information. The route request (RREQ) packets and route response (RREP) packets are broadcasted using omnidirectional transmission. Whereas, the data packets are exchanged using a directional antenna, but, if the node doesn't receive ACKs, then it uses omnidirectional transmission mode.

The authors of [136] combines omnidirectional and directional transmissions for UAV emergency communication system. The system architecture is composed of a lot of UAVs, a mobile base station and a satellite transmission access. The base station is connected to the mobile communication network via satellite transmission using omnidirectional and directional antennas. The Ku-band satellite allows the transmission access using the airborne satellite communication equipment. The proposed system conducted to solve some emergency communication problems: (i) the flight altitude can reach 200m which is restricted to other emergency communications vehicle, (ii) the damaged roads when the flood or the earthquake occurs.

6.2.2. Classification based on access mechanisms

In this section we will discuss several MAC protocols using the following classification: 1) CDMA-based protocols 2) TDMA-based protocols 3) NOMA-based protocols 4) Hybrid protocols, and 5) cross-layer protocols.

A. CDMA-based protocols

In [137] authors studied the MAC design in the UAV networks using full-duplex radios and multi-packet reception (MPR). To improve the MAC performance, they used token-based technique and they distinguish two cases: the case of the presence of perfect Channel

State Information (CSI) and the case of the presence of imperfect CSI. In the first case they proposed an approach for transmission optimization and in the second case, they proposed an approach for discrete stochastic optimization. Simulation results show that the proposed scheme outperforms the existing based token scheme and the existing based token and MPR capability scheme by maximizing the network throughput and minimizing the delay. In [138] authors proposed a token circulation scheme for CDMA-based UAVs networks. In this scheme a token continuously circulates in the network, consequently, each UAV can detect its unknown and lost neighbors in real-time. Every UAV assigns codes with reusing of spatial channel codes without generating code collision and it schedule data transmissions in a distributed and cooperative manner. Analytical and simulation results show that the latency of this scheme is relative to the network size. Besides, the average delay of the data packet is proportional to the packet generation rate and the network size.

B. TDMA-based protocols

In [139] authors implemented UAVs communication protocol using TDMA based on slot ALOHA. This protocol achieves the UAVs real-time communication and it generates a low collision rate and ensures high levels of channel utilization. In [140] authors studied the delay of UAV data link basing on TDMA and Aloha. Specifically, they considered a data-link semi-physical simulation test platform. They discussed the UAV average delay under different MAC protocols using a darkroom calibrate-link method. The simulation results show that when the TDMA protocol is used for short-range UAVs, the success of receiving data packets from UAVs to the GCS is independent of the UAV network size. Whereas when the TDMA protocol is used for long-range UAVs, the average of receiving data packets from UAVs to the GCS will decrease slightly with the increase of the UAV network size.

Moreover, in [141] the authors proposed a multi-user TDMA-based Direct Sequence Spread Spectrum (DSSS) system in which multiple users get their messages simultaneously without any interference. The proposed system covers multiple users in one slot by designating orthogonal codes to users. The Barker code is used for spreading data with lengths of 7, 11, and 13. Simulation results show the importance of the factor of code length to reduce the noise power which decreases the error probability. To improve system performance, the Hadamard code should be used to increase the number of users per slot.

Authors of [142] used UAVs as relays between the ground and the satellites. They considered a two-level relay structure based on TDMA. Users transmit packets to the UAVs via a random access scheme. Then, the UAVs forward them to the satellite with a coordinated access scheme. The coordinated access can substantially enhance the delay and the throughput performance compared to the uncoordinated access.

C. NOMA-based protocols

The Non-Orthogonal Multiple Access (NOMA) consist of serving multiple users on a unique resource block (space/ time/ frequency/ code), and with different power levels. At the transmitters, NOMA assigns the power resources to all users based on the Superposition Coding (SC). At the receivers, the Successive Interference Cancellation (SIC) is used to exploit the channel for the different users. Therefore, NOMA enhances the connectivity and the meeting of diverse user requirements. In [143] authors proposed a novel framework for UAV networks based on NOMA. It evaluated the NOMA performance for UAV networks by using stochastic geometry for the UAV's positions and the ground users. The SIC decoding characteristics of NOMA cause considerable delays if the NOMA users' number is large.

In [144] researchers supposed that the traditional NOMA with only local SIC provides very limited gain comparing to the Orthogonal Multiple Access (OMA) because of the presence of a large number of

co-channel. To mitigate the UAV uplink interference, they proposed a cooperative NOMA scheme based on the existing backhaul links among BSs. The proposed scheme increases the weighted sum-rate of the ground users and the UAV by jointly enhancing the UAV power allocations and rate over multiple resource blocks. Another research work [145] addressed the interference problem in NOMA based UAV satellite networks. The authors supposed that the power allocation is important to NOMA because it influences the SIC performance and the achievable rate of users. They proposed a power allocation algorithm to minimize the maximum users' outage probability. Simulation results confirmed the convergence of the power allocation algorithm.

In [146] authors proposed a multi-beam transmission method by exploiting the NOMA technique. Specifically, the UAV sends each data to a chosen subset of the BSs. Each BS can decode the UAV signals and cancels them before decoding the users' messages to null its interference at the other BSs. Therefore, the proposed method allows us to achieve a high transmission rate in the cellular network and avoid interfering with any existing communications at the co-channel BSs.

D. Hybrid protocols

In [121] authors proposed a Collision-Free MAC protocol (CF-MAC) with omnidirectional antennas and half-duplex radio. When a node would connect to the channel, it will get one free slot and its access with CSMA backoff method. This protocol enables the UAVs to access rapidly to the networks and it uses a Region Marking (RM) scheme to decrease the collision probability and eliminate the hidden terminal problem. Authors compared their protocol with VeMAC protocol which is a MAC protocol for VANETs. Simulation results show that CF-MAC improves the efficiency of channel access and it decreases the collision probability. Moreover, in [147] researchers proposed a novel hybrid protocol for data gathering in a UAV-WSN. It combines TDMA and CSMA/CA for the channel access. The UAV sends periodically a beacon frame to the sensor nodes to inform its presence. Each sensor node receiving the beacon frame responds with a registration frame. The UAV sends a second beacon frame to the registered nodes based on their transmission schedule. The proposed protocol uses CSMA/CA for the registration process and allocates the time slots of the registered sensor nodes in the data gathering process. Then, the time-slot scheme is exploited to determine the sensor transmission schedule based on the priority in the registration process.

E. Cross-layer protocols

In [83] the authors proposed a cross-layer approach to adjust several key parameters in the initial three layers basing on the aerial attitude variations. To increase the number of hops and extend the coverage area, the authors used a directional antenna. In this work, two schemes are combined: Intelligent MAC protocol (IMAC_{UAV}) and the aforementioned DOLSR protocol. In the IMAC_{UAV} scheme, packets coming from the upper layer trigger the transmission of the control RTS packets which include the position of the UAV and the duration of the transmission. RTS and CTS packets are transmitted through the omnidirectional antenna. Every UAV which hears the CTS and RTS control packet must store this information and modify its table. The data packet will be sent using a directional antenna. The authors compared their scheme with the IEEE802.11 standard. Simulation results show that this scheme gave a better end-to-end delay. In [148] the authors proposed an opportunistic cross-layer protocol for efficient data dissemination over UAVs networks. Tasks to be executed in the UAVs networks are planned into series of services. These services have a layered structure and use the available resources of each node for dissemination.

Based on the studied UAVs MAC protocols, we have identified the advantages, the suitability, and the limitations of each MAC protocol. The results of these investigations are regrouped and highlighted in

Table 6
A comparative study between UAVs MAC protocols-based antenna categories.

		Basic Concepts	Suitability to UAVs	Advantages	Limitations
Omni Directional Antenna	[121]	Region marking scheme	-	Reduce the collision probability channel access	Cumulative estimation errors
	[122]	Two broadside omnidirectional antenna	-	Easy in the installation Generate significant network gain Higher access accuracy	Increase interference
Directional Antenna	Fixed beam	[123] Antenna arrays are controlled using an integrated feed network	+		Low antenna gain
	Switched beam	[124] Rotate the UAV to point a beam in the desired direction [125] UAV is equipped with a pair of transceivers	+	Improve the system capacity Improve the spatial reuse network capacity	Increase the UAV rotation time Increase useless packets
Adaptive Antenna Array Systems	[128]	Backhaul Link with Antenna Steering Keep the antenna steered toward the GS	+	Efficient signal strength	Unreliable communication system Low antenna gain
	[129]	Field Programmable Gate Arrays Fast Fourier Transform	+	Low complexity Improve the system gain	Increase hardware requirements
Hybrid	[130]	Two beamforming and scheduling scheme	+	Minimize the MSE Improve throughput	Increase transmitted packets
	[131]	Diffuse signal to every receiving antenna	+	Find the Direction Of Arrival More received data packet	Increase the use of the bandwidth
	[132]	Wide beam antenna array for U2X Two dipole antennas arrays with parasitic elements	+	High gain Higher access efficiency	Not suitable for multihops applications Energy inefficiency
	[133]	Measure the distance between UAVs Check the UAV altitude	+	Improve the network performance	Increase delay
	[134]	Transmissions based on location and trajectory information RREQ and RREP packets	+	Higher access accuracy Low link interruption	Complex system design Not suitable for dense network
	[135] [136]	Utilize location and trajectory information Hybrid antennas for BS Satellite transmission access	+	Decrease packet loss rate Reliable communication system	Decrease the throughput Low signal strength due to the UAVs high-speed

Table 7
A comparative study of UAVs MAC protocols-based access mechanisms.

		Basic Concepts	Suitability to UAVs	Advantages	Limitations
CDMA-based	[137]	Full duplex-radio	-	Maximize throughput	Based on channel state information
	[138]	Multi-packet reception Token circulate continuously in the network	-	Minimize delay Each UAV detect unknown and lost neighbors	Latency relative to the network size
TDMA-based	[139]	Slot Aloha Real-time communication	-	Decrease collision rate	Increase delay
	[140]	Darkroom Calibrate-Link method	-	High level of channel utilization Decrease average delay	Received data packet rate is relative to the network size
	[141]	Cover multiple users in a unique slot	-	Designate orthogonal codes to users	Few number of users per slot
	[142]	Use Barker code	-	Reduce the noise power	No consideration to the frequent topology change
NOMA-based	[143]	Ground-to-UAV and UAV-to-satellite links	+	Delay and throughput enhancement	Introduce considerable delays
	[144]	Coordinated access scheme	+	Efficient for frequent network topology variation	Require excessive power for signal processing
	[145]	Use of stochastic geometry for the UAVs positions	+	Increase the weighted sum-rate	No consideration of the UAV mobility
	[146]	Cooperative NOMA scheme	+	Mitigate the uplink interference	Presence of inter-cell interference
Hybrid	[147]	UAV sends its data to subset of BSs	+	Allow to achieve high transmission rate	Increase delay
	[148]	BS cancel the UAV signals before decoding the packets of its user	+	Efficient channel access	No consideration of the optimal flight path
Cross-layer	[149]	Multiple satellites cooperatively serve the UAVs	+	Minimize the outage probability	energy consumption
	[150]	Power allocation algorithm	+	Guarantee the user fairness	Decrease data packet rate
	[151]	CSMA backoff method	+	Rapid access to the network	Decrease throughput
	[152]	Region Marking scheme	+	Efficient channel access	Increase collisions
	[153]	Combine TDMA and CSMA/CA	+	Send of UAV beacon frame to the sensors	
	[154]	to inform its presence	+	Adjust parameters in three layers	
	[155]	RTS packets contain UAV positions	+	Extend coverage area	
	[156]	Use the available resources	+	Decrease delay	

Tables 6 and 7.

7. UAVs mobility models

Mobility is an important factor in the communication protocols design for UAVs networks. In fact, in the evaluation of the developed protocols, it is required to use correct mobility models for particular applications to have a deeper analysis of the performance. Mobility patterns precisely capture the movement behavior of each mobile node and it specifies their movement models that enable a deep statistical analysis with different performance measures, like end-to-end delay, packet delivery ratio, throughput, and overhead. In the literature there are many mobility models, some of them are designed for VANETs networks [149] and it is adapted to the UAVs networks and others are designed specifically for UAVs networks. Those different mobility models can be classified into five categories 1) pure random models, 2) models with temporal dependency, 3) models with spatial dependency, 4) models with geographical dependency, and 5) hybrid models.

7.1. Pure random models

It includes the Random Walk (RW), Random Waypoint (RWP), and Random Direction (RD) mobility models.

7.1.1. Random walk model (RW)

Random Walk model specifies the movement of UAV by the random variables which change the direction when the UAV moves in this direction. At boundaries, UAV returns back from the boundary or it can wrap around to the other region side [150]. This model has two methods: time method and distance method. The time method estimates how long the UAV will keep its current velocity and direction before selecting the new values. The distance method defines the time before choosing the new values of the movement basing on the current speed and the distance.

7.1.2. Random way point model (RWP)

The Random Waypoint model considers UAV flying to a destination chosen randomly in a region with a randomly chosen speed. It can pause for a short period between different destinations. RWP and RW have similar trajectories in terms of directional changes. The major difference is that nodes of the RWP model may appear more frequently near the center of the region. There are many researches which are focused on studying the RWP statistical properties, such as node distribution [151,152], transition length [151], and link and path durations [153]. In the RWP model, the UAVs distributions are non-uniform, even if there is a uniform initial distribution. This non-uniform distribution is produced because of the region boundaries restriction and tacking nodes toward the region center.

7.1.3. Random direction model (RD)

The RD model [150] is developed to address the non-uniform distribution of nodes of the RWP model. In the RD model, each UAV chooses randomly a direction and speed, may fly to the boundary, stop for a while, and after that randomly chooses the new direction to move. The RD and RW trajectories models are similar, the only difference is in the traveling duration: constant or random. In the RD model, the stationary node distribution is uniform, in spite of the initial distribution of nodes. In the aforementioned models, the high mobility is easily captured, by configuring a variable speed. Besides, these models are simple to implement, it doesn't need a message exchange mechanism, and so, it doesn't require a network protocol. Whereas, when there is a heading change, the novel heading is uncorrelated to the current heading. Moreover, these models don't register the mobility information of the different randomly-selected time slots. Random models are abstract and unrealistic for the UAVs networks because it ignores the spatial and temporal correlation and geographical constraints and

doesn't mimic aerodynamic constraints UAVs.

7.2. Models with temporal dependency

A Gauss-Markov mobility model [154] is a temporally correlated mobility model. It avoids sharp motion changes. This model is used for aerial target tracking. The continuous-time dynamics let this model more suitable UAVs networks and the simulated area size is variable. UAVs near boundaries can wrap around to the other region side which is similar to the RD model. In order to limit the sharp direction change in boundaries which is frequent in the 2D concept, this model is extended to 3D UAV position is always determined by its previous position because of the high moving speed. The model memory contains the path of each UAV. In this model, each UAV is initialized with direction and speed. After every fixed interval of time movement, we have to update the direction and speed of each UAV.

7.3. Models with spatial dependency

7.3.1. Smooth turn mobility model (ST)

The ST mobility model [155] developed to capture the freely-moving UAVs to make smooth trajectories. In the basic 2D Smooth turn mobility model, UAVs choose a point from the 2D plane which is in the perpendicular line to its heading direction. The perpendicularity guarantees the smooth turning trajectories. The behavior of vehicles on the boundaries is modeled by reflecting back to the region, or returning and appearing at the other part of the region. UAVs can, also, use a realistic boundary model like the model used in the Three-Way mobility model. The 2D ST mobility model is extended to 3D. In the 3D ST two mobility models are developed: the z-dependent and the z-independent. In the z-dependent ST, planes are introduced to capture the movements along the z dimension. The z-dependent ST uses the same movement in the plane as in 2D ST. The difference is in the time of change in the turn center; each UAV randomly selects a plane and chooses a new turn center in the plane. The newly selected plane intersects the current plane with a line presenting the current heading direction. The movement of the z coordinate, in the z-independent ST, is independent of others in the x and y plane. The movement in the z dimension is modeled separately to reflect existing flight data. The ST model ensures smooth turns with flexible radii. It captures frequent network topology changes and high mobility. In this model, UAVs are free to travel in the simulation area with changeable turn centers and radii.

7.3.2. Mission plan based mobility model

This type of mobility model [156] uses pre-defined trajectory plans. The initial flight plans are checked in the global mobility file that contains the location information of each UAV. After that, the global mobility file is used to explore the map of the Time-Dependent Network Topology (TDNT). Whereas, uncertainties in the environment can cause a modification in the flight plan, therefore, hello-and-acknowledge mechanisms are used to check if the TDNT is up-to-date. Flight plans are typically used because of the safety of UAVs, especially in autonomous flying. This model captures the high mobility, aerodynamics, and safe constraint, as all the UAVs are reflected by mission plans.

7.4. Hybrids models

7.4.1. Three-way random and pheromone repel mobility model

Three-Way Random mobility model [157,158] assumes that the heading speed and the turn radius are constants at all times. This model defined the mobility pattern on a Markov chain, and it uses three mobility modes: going straight, turning left, and turning right. The selection of mobility mode of the next time is dependent on the current mode, with the conditional probability. The movements close to boundaries are similar to the Gauss-Markov model; so, if a vehicle is at a certain distance from the boundary, then it selects a turning direction

away from the boundary. The Pheromone Repel mobility model is developed based on the Three-Way Random model; it permits each UAV to adjust its direction to improve scan coverage, by avoiding areas which are recently visited. This model uses pheromone maps to select mobility modes. The Pheromone Repel model enhanced coverage properties by comparing to the Three-Way Random model. Whereas, Pheromone Repel model has a much number of disconnected clusters by comparing to the Three-Way Random models in transient time. Both the Pheromone Repel models and the Three-Way Random allow UAVs to perform typical turns using constant turn rates. These two models, unlike the SRCM model, do not need a fixed turn center.

7.4.2. Semi-random circular movement model (SRCM)

SRCM [159] is a specific mobility model for UAVs networks. This mobility model limits the movement of the UAVs to circulate around a fixed center. This model is used in scenarios where the target location is known, so, UAVs are dispatched to accumulate information in the nearby area. The SRCM model generates less variation in connectivity probability that ensures more stability in a communication network. Moreover, this model assures smooth turning trajectories limited by aerodynamic and mechanical constraints. Besides, by comparing this model to the RWP model, this model has quicker coverage speed and a higher steady-state coverage percentage. Whereas this model assumes that the circling time is much high than the transition time, it neglects the non-smooth movement during transitioning.

7.4.3. Multi-tier mobility model

The airspace contains heterogeneous flying devices with different types operating for different missions [160]. Therefore, networking in such heterogeneous networks needs mobility models that incorporate multiple mobility patterns. The multi-Tier mobility model contains different types of UAVs flying at different altitudes. UAVs networks can be connected to fixed control stations to make multi-domain communication networks. Authors of [161] presented a new 3D design of the Gauss-Markov mobility model for multi-tier, and dynamic MANET-airborne networks. They assigned for each mobile node an initial direction and speed. After an interval of time, the new direction and speed of each node are calculated, in order to follow the new course. This cycle is repeated through the simulation duration.

8. Security challenges of UAV networks

Securing the closed-loop of the UAV networks a challenging task because they are generally deployed in unstructured environments and terrains, and any security problem in each level will paralyze the whole system. Typically, different types of cyberattacks (jamming, hijacking, eavesdropping, disturbing, injecting wrong data, etc) can be occurred in the UAV networks. From the CPS perspective, UAV networks represent the integration of (i) sensing, (ii) communication, (iii) computation, and (iv) control. In the following, we will discuss the essential UAV network security requirements and the related solutions at each component.

8.1. Sensing

UAVs are equipped with several remote sensing devices to introduce the data from the physical world into the UAV network. Sensor spoofing attacks are steered towards on-board sensors such as the GPS receivers, vision, lidar, radar, sonar, and IR sensors. It can cause mode confusion and overriding gains through hacking. The main intentional vulnerability is GPS spoofing which is a major menace to the civilian GPS. In the GPS spoofing attacks, the attacker tries to transmit false GPS signals to orient the UAV to a false location without disrupting the GPS operation.

Authors of [162] proposed a vision-based spoofing detection method. It is based on the comparison of the UAV relative sub-

trajectory from Visual Odometry (VO), with the replica from GPS in a moving window during the flight path. This method can detect the time and location of GPS spoofing. It can be exploited without any requirement for the usage environment.

Authors of [163] a UAV GPS spoofing detection algorithm based on the fusion of GPS and optical flow raw data. The fusion model is compared to the fusion data model in the normal state. The difference between the two models is used to detect whether the sensor is attacked. Authors of [164] proposed a machine learning method to determine GPS spoofing signals based on the artificial neural network. They used different features like pseudo-range, signal-to-noise ratio (SNR), and Doppler shift to perform the GPS signals classification.

8.2. Communication

Communication guarantees the data distribution and sharing. The wireless links, the uncontrolled environment, the dynamic topology features make the UAV networks vulnerable several types of communication attacks. Some of the attacks possible in UAV networks are Man-in-the-Middle (MIM), Injection, Sniffing, and Denial of Service. The integration of BlockChain (BC) technology over the communication channels can be a good solution which is exploited in many research works. BC can provide solutions for authentication, integrity, availability, and confidentiality security issues in UAV networks.

1. Authentication

It is important to authenticate all nodes and packets which transit within UAV networks. Only authenticated nodes can collaborate in the routing process. Generally, there are two types of authentication services: Node authentication and Message Authentication. Node authentication guarantees that only authorized GCS and UAVs can access to the inherent network information and the resource. Message authentication allows to authenticate the message and to ensure the message integrity. Authors of [165] proposed an authentication framework for the IoD. They used a predictive authentication approach. They considered that identity, location, and flying routes of each UAV are important information in the IoD network. However, the proposed scheme did not consider mutual authentication to ensure the communication entity. The mutual authentication issue is addressed in [166], in which authors integrated the elliptic curve cryptography, hash function, digital signature, and other cryptography algorithms for UAV application. Authors of [167] introduced a BC-based access control algorithm in the internet of drones environment to permit secure communication between the drones, and between the drones and the GS. Secure data collected by the GS compose transactions which are made into blocks. Finally, the blocks are added to the blockchain. Once the addition is made, the transactions can not be modified, altered or even removed.

In WiFi Attack, the attacker tries to intercept the UAV's data messages over WiFi channels. In this context, authors of [168] studied the cyber-security issues in UAV-assisted public safety networks. They proposed attribute-based encryption for the management and also homomorphic encryption for the data aggregation. The proposed scheme can protect UAV networks against WiFi attacks.

2. Integrity:

It permits the no interception of the exchanged messages between the GCS and the UAVs. It ensures that an adversary cannot intervene by insertion, deletion, or modification of messages. Neighbor nodes need to guarantee that an integrity mechanism is launched to detect adversary nodes.

Authors of [169] proposed a BC-based data gathering using UAV. UAV securely keeps data in blockchain at mobile edge computing

server. Data are encrypted before being transferred to the server. After receiving the data, the mobile edge computing server validates the sender identity and the data. Once validated, the data will be stocked into the blockchain.

Authors of [170] proposed a BC-based trust management mechanism to reduce network latency and overheads. All transactions are registered on the BC and disseminated to all participating UAVs by the parent or coordinator, or controller node to maintain the integrity in the system.

3. Availability:

It ensures that all services given by the UAV networks are always available. The network is useful and functional. Data such as command and control traffic should be always available. Authors of [171] proposed a network architecture based on the Dyna-Q scheme to avoid interference attacks. The communication solution uses dual controllers, including the software-defined network controller, and the UAV flight controller. These dual controllers collect the state information of the network that is used by the reinforcement learning model.

Another availability attack of the UAV networks is the jamming in which the adversary can disrupt the communication by adding interference on the receiver nodes. In this context, the authors of [172] proposed a device jamming attack detection based on deep-learning. They added a client group prioritization mechanism to a federated learning model. The client group prioritization technique allows the aggregator UAV to have a good identification of client groups to calculate the global update. Moreover, authors of [173] proposed a UAV-enabled jamming noise mechanism for securing communication in cognitive radio networks. They used a secondary transmitter to send confidential messages to another secondary receiver. The UAV acts as a jammer to degrade the capability of the external eavesdropper. Therefore, the resource allocation must jointly optimize the UAVs trajectory and the transmission power with satisfying an interference at the primary receiver.

4. Confidentiality:

It prevents unauthorized users to access to the nodes' conversation. Not only UAVs and communication links are vulnerable to this attack but also GCS. The most frequent attack is Man in the Middle (MIM). In the MIM attack, an attacker controls or modifies the messages between the sides. Therefore, it can pass false information or provide false commands between the UAVs and their ground stations. In this context, authors of [174] proposed a mechanism to maintain the security of the UAV network based on BC by verifying information of the conversations from different sources. The proposed technique monitors the behavior and UAVs positions like illegal border crossings. Besides, it uses asymmetric encryption with some official UAVs to decrease the MIM attack.

Eavesdropping attack means that an attacker can secretly listen to the communication between the UAVs. This attack is also known as sniffing or snooping attacks. It is a passive form of the MIM attack because it does not cause any network transmission. Authors of [175] considered a mechanism for preventing eavesdropping attacks in a UAV network. The UAV is used as a relay and an uplink phase for authentication. To detect the eavesdropping attacks, they considered building predictive models with K-means clustering and support vector machines. To do, they proposed two frameworks: (i) The first framework creates the features of checking data from wireless signals, and (ii) The second framework generates the training data.

8.3. Computation

Computation is exploited for analyzing and decision making using

all the acquired data. UAVs have limited storage and computing capacities. An attacker can launch the sleep deprivation attack that tries to drain UAV battery. Moreover, an attacker can produce a flooding attack to exhaust the network bandwidth and consume UAVs and GCS resources. Authors of [176] proposed a real-time protection method against denial of service attacks. They prevent nodes to access intensively to the memory bandwidth.

Authors of [177] proposed an energy-efficient and low-latency computation offloading method for UAV mobile edge computing systems. To do, they formulated a set of energy-efficiency issues in the presence of both passive and active eavesdroppers. Their optimal solutions are analyzed from a physical perspective.

8.4. Control

Control translates the decisions to instructions for the actuators of the physical world. An attacker try to take control of the UAV to enable the injection of the desired values. Authors of [178] proposed an online detection method of the faults UAVs actuators. A neural network is exploited for the fault detection, and its weighting factors are updated by an Extended Kalman filter. The weighting factors adaptation permits to detect intermittent faults accurately. Moreover, researchers of [179] proposed three solutions to detect and mitigate the actuators attacks of UAVs. The first method consists of a detection strategy by designing excitation signals to the control commands to enhance the detectability of the attack. The second method designed an unknown input observer which detects the attack and also estimates the attack magnitude. The third method designed an actuator system that uses different modulated signals with variable frequency pulse-width to prevent the actuator against malicious attacks.

9. UAVs localization methods

Global Positioning System (GPS) and Inertial Navigation System (INS) are the most common systems used in traditional aerial vehicles to ensure the localization. Whereas INS and GPS have serious disadvantages, for instance, INS can generate accumulated errors and GPS presents usually high satellite signal loss. Therefore, it is required to use an alternative supporter system. Authors in [180] proposed to include a Terrain Referenced Navigation (TRN) in the localization algorithm for UAVs. They initially discussed the basic terrain reference model and they used standard altimeters in the analysis. They used the binary searching for the UAVs trajectory control system and the estimated path is computed for their flight. To improve the computation it is required to store data that require a large memory space especially in unknown and complex terrains having more geographical barriers. Every UAV participating in the localization process prepares a dataset basing on its identification process and storage profile.

In [181] researchers proposed an algorithm for cooperative UAVs based on the decisiveness and control. They used a randomized examination for UAVs based on tree data structures. The UAV localization is determined by using these tree configurations. The algorithm of the cooperative UAVs is divided into three layers: the control layer, the communication layer, and the formulation layer. The control layer is based on the trajectory control which ensures the solidarity, the alignment and the variance between UAVs. The objective formulation layer identifies the targets using flocking algorithms. The communication layer between the vehicles is based on rooted spanning tree the configuration. In [182] authors developed robust techniques for multiple cooperative UAVs for searching, detecting, and locating mobile ground targets sending radio frequency signals. To do so they used combining sensor data from cooperative multiple UAVs to get accurate and reliable target locations. They applied Kalman filtering techniques

to the collected data to estimate the angle to the target. Then, the estimated angles of the target from the multiple UAVs permit to collect additional measurements for the control mechanism which converge to the estimation of the targets position.

In [183] authors used a navigation control scheme and radio frequency (RF) detection for localization. The RF detection uses a directional antenna and UAVs fly to determine the bearing of the GPS jammer. On the other hand, the navigation control scheme chooses the optimized trajectory for the rapid localization of the jammer. In [184] authors created a 3D localization mechanism using the technique of Angle of Arrival (AoA), combined with Wi-Fi technology and directional antennas. The research work aims to locate a UAV that is driven by a smart network using directional antennas in indoor environments. The UAV distributes a Wi-Fi signal to the connected sensing points. To find the location of the UAV, the antennas determine the RSSI values and calculate the angles. With this information, lines are traced between the UAV and the antennas. Therefore, with the intersection lines, it is possible to indicate the position of the UAV.

In [185] the authors implemented a localization and attitude detection module for UAVs by incorporating sensing data from the GPS, IMU values, and digital compass. The authors used the Arduino platform as the processing unit to calculate the combined attitude and localization information. Then, this information is sent to the ground station in real-time. Besides, they used the zero velocity updated method to detect the vehicle motion from IMU measurements to update the localization information. In [186] the authors implemented UAV localization algorithms for a post-earthquake monitoring system. To do so, they installed a Light Detection and Ranging Sensor (LiDAR) and a high-resolution camera on the UAV to obtain detailed topological maps of the physical surfaces using vision and marker localization methods. The camera is able to measure the position and the orientation of the UAV. They used Bayes recursive filters to properly represent the uncertainty inherent in the UAV estimation.

10. QoS provisioning for UAVs networks

In the UAV environment, there are many mechanisms needing to configure and assure QoS requirements. For example, UAV could be sending critical data with certain error levels and time frames. Moreover, the QoS requirements can be in multiple considerations such as collaborative sensing and safety. In the UAV to UAV communication, UAVs use a multi-hop routing protocol, a specific path must be selected which meets certain QoS requirements like bandwidth, end-to-end delay, jitter, packet loss, etc. Therefore, defining a comprehensive framework enabling QoS at the middleware is a critical challenge.

In [187] authors proposed a QoS control scheme basing on the process-patterns considering the dynamic behavior of UAVs. They used a hierarchical communication structure which is based on every UAVs aerial position and it is controlled by the QoS scheme. The QoS management consists of two components which are Queue Manager and Process-Pattern Manager. Two different modes can be used: Standard mode and SuperNode mode. In the Standard mode the Queue Manager must queue its own forwarding traffic. In the SuperNode mode, UAVs are utilized as SuperNodes, and the Queue Manager also must handle the outgoing traffic. The queuing is executed according to the information given by the Process-Pattern Manager. This last manages the existing process-patterns then identifies the active service task. Therefore, the scheme optimizes the communication paths between a central controller and UAVs, and it works closer to real QoS requirements. In [188] authors proposed a novel QoS scheduling scheme which allows maximum flexibility in the parameterization of the multiple service classes. The scheduling algorithm is used to guarantee that the UAV control data are transmitted under all circumstances in a very simple

and flexible way.

In [189] authors proposed a QoS-aware access channel management scheme for vehicular networks. The proposed mechanism can be used in adaptive and QoS-aware routing protocols for UAV systems using cognitive radio technology. In [190] authors analyzed the transmission efficiency and the required bandwidth to guarantee the QoS in densely distributed UAVs connected with the randomly placed GCS. They are focused on the received Signal Interference Noise Ratio (SINR). Simulation results show that the values of the received SINR depend on the antenna directivity, therefore, the required bandwidth can decrease with a sharp beam pattern. However increasing the transmit power, improves the transmission efficiency slightly because of the limited interference region of the highly distributed UAV. Moreover, when the GS is placed in a random location, the larger bandwidth is needed to guarantee the QoS. In [191] authors defined a suitable UAV network architecture to achieve the QoS level necessary. They also studied the QoS communication to provide safe communication over UAVs networks by introducing design methodologies and evaluations of some routing protocols of the UAVs network in the real and simulated environment.

11. UAVs applications

Each UAV application domain differs in its system requirements and that can affect the network parameters varying in frequency, delay tolerance, traffic type, communication range, energy limitations, etc. However, nearly the coordination and path planning are the common requirements for all the multi-UAVs applications to realize the mission goal and operate on a high level. On the one hand, the coordination mechanisms must provide the following: attitude alignment, team formation, coordinated decision making, coupled oscillators, position synchronization. For example, in [192] authors presented a decentralized optimization of UAV swarm control which offers methods to coordinate UAVs with multiple decision-makers. Moreover, the proposed model takes into account the data communication in the system including the messages delivery uncertainty.

On the other hand, UAVs operating in a diversified environment containing various obstacles must take into consideration the path planning process. The path planning algorithms must produce the solution, in a very short time. In [193] authors proposed an algorithm to optimize the path planning of the UAVs depending on the computation time. The UAVs are considered having limited-communication and limited-sensor capabilities. Each UAV estimates its path planning, and then it distributes its plan with others if the communication distance allows. During the flight, UAVs sharing information can update their paths to avoid the collisions.

In [194] authors proposed adaptive algorithms for path planning for UAVs cooperatives networks. In the proposed solution, the exceptional events are tolerated, such as the leaves/joins of the UAVs from/to the network, the enters/leaves of the sensors to/from the network. Moreover, UAVs can communicate with each other; therefore, the path is computed collaboratively. The algorithms of the collision avoidance are usually implemented in networks containing cooperative swarms of UAVs, since UAVs may operate in the close area, and during the operation, some of the vehicles may need to change their position.

In [195] authors presented a comprehensive collision avoidance study for various types of vehicles. They are focalized on the distributed application for UAVs networks considering that UAVs share their location and estimated direction of movement between each other. This information is updated by the communication algorithms; therefore, its ready even in critical moments. The Collision Solver Manager (CSM) [12] unit is employed to manage both the collision detection and the collision solving; it uses the results from other collision solvers to

generate the final avoidance solution.

UAVs applications include, but not limited to, data collection, infrastructure inspection, construction, area surveillance, search and rescue, target tracking, healthcare, and delivery and transportation.

11.1. Data collection

UAVs are used to gather data from WSN sensors in the disaster areas, desert areas, mountain areas, and so on. In the literature, there are data collection algorithms based only on the central unit and there are others which treat and merge data firstly in UAVs. In [27] the authors presented the automated sensing platform using multiple UAVs. The data gathered is examined at UAVs, to manage the noise introduced in the UAV operation. Then the gathered data is merged in a cooperative distributed manner at UAVs. In [196] authors solve the problem of centralized data fusion. Therefore, there is no common communication unit; the fusion of the data is established in a distributed cooperative manner at UAVs, whereas, sensors just know the neighboring network connections. The decentralization process is realized using the Kalman filter which requires the fully connected network, but in practice can be realized only in small-sized systems.

11.2. Infrastructure inspection

The researchers of [197] proposed a UAV system for infrastructure inspection automation. The proposed system controls the UAV towards some inspection points which are chosen by the operators. It allows us to inspect unreachable areas in seconds. To select the best area to observe the target points a planning algorithm is used. The planning algorithm should also calculate safe trajectories for the UAV.

The authors of [198] proposed a UAV goal-aware navigation algorithm for infrastructure inspection. The algorithm estimates the UAV orientation and the related landmarks with respect to the goal. The evaluation results show the robustness of the algorithm even in the presence of similar-looking or similar-colored object.

The researchers of [199] proposed a railway infrastructure monitoring algorithm for UAV based on computer vision. The UAV takes images of the railway infrastructure and the rail tracks by means of a camera. The edge and the feature extraction methods are applied to these images to determine the rails. Employing UAVs for such inspection systems permits more robust visual monitoring while providing a low cost.

11.3. Construction

UAVs have presented the potential to enhance the construction safety performance thanks to their high-resolution camera-equipped. In the literature, there are two approaches for using UAV in construction safety. In the first approach, UAVs are used to identify potential safety issues. In the second approach, experts use pictures taken by UAVs to identify the potential safety problems and generate UAVs flight routes.

Adopting the first approach, authors of [200] a framework utilizing swarm intelligence to automatically produce the UAV inspection plans. The objectives of inspection, the geometry, and the semantics are automatically extracted from the building information models. Then, based on artificial intelligence, the optimal inspection plan is generated ensuring complete inspection coverage with minimizing the flight duration.

Authors of [201] used images taken by a UAV to show the present stage of construction that will be used to design the safety system. Particularly, the UAV surveyed an educational building which is under construction. The related safety analysis was studied based on the pictures acquired from the UAV.

Adopting the second approach, authors of [202] investigated the UAV aerial visual contents for conducting the safety planning of a high-rise building construction. Based on the aerial videos captured by the UAV, experts identified the hazards of unsafe acts and conditions at height like safety nets around unprotected edges, missing guardrails, and unsecured material.

Researchers of [203] proposed UAV-assisted visual localization and monitoring algorithms for measuring the proximities of the construction entities. The object localization algorithm is developed based on a deep neural network to provide fast and robust localization entities. The image rectification method measures the actual proximity from the images collected from a UAV. When executed together, the algorithms can automatically monitor the proximities of the construction entities.

11.4. Area surveillance

Area surveillance applications operated by a team of UAVs are growing frequently. To perform the surveillance task through the team of heterogeneous UAVs equipped with various sensors it is required to use dynamic task reconfiguration. Moreover, multiple tasks require large computing power so increasing the use of the processing resources brings a valuable profit.

In [25] authors proposed an approach of the distributed cooperative perimeter surveillance application. The proposed approach uses the team of UAVs and tolerates the insertion/deletion of the UAVs team. The results are transmitted to the ground station which constitutes the system central element. UAVs communicate just with their neighbors that are in their coverage area which is another problem in the distributed processing. Moreover, the distributed perimeter surveillance application must take into consideration the prioritization of area sections.

11.5. Search and rescue (SAR)

SAR operations can occur in natural disasters, accidents, finding missing persons, evacuation process to give immediate help, etc. Such operations are time-critical because it is needed to be achieved in the shortest time. UAVs are suitable for such operations thanks to their flexibility of deploying.

Authors of [204] proposed a SAR method for the swarm of UAVs using a genetic algorithm. The UAVs are able to self-organize the mesh network which is optimized for area coverage. They are able to know the communication strength of all the connected UAVs using RSSI, and adjust their placement closer to other UAVs when RSSI is below a threshold. They can also maintain communication as a group when they move together in a search path.

Researchers of [205] considered the use of UAVs to accomplish a SAR operation in the minimum time while rescuing the maximum number of victims. The proposed technique simulates real disasters task-distribution among UAVs. This technique is compared with some bio-inspired and opportunistic task allocation schemes, in terms of the number of rescued victims and the execution times. The simulation results show the high performance of the proposed technique.

11.6. Target tracking

The object detection [206] [229] is an application of UAV swarms implementing distributed processing, usually based on the gathered data from data sources such as radars and cameras. In [207] authors presented the use of Bayesian search and the localization in a dispersed manner. Both the search and the localization algorithm use the method of filtering multiple observations arriving from the cameras attached to vehicles. In the case of single target localization, the probability of

successful detection of a target is based on the image resolution and the target visibility. The multiple-target search is presented in [208]. Each one in the team of UAVs is equipped with a camera. The authors divided the monitoring area into regions, and they calculated the probability of target location in every one of the regions. The distributed UAV platform utilizes the sharing of measurement data, information may update due to the changes in the environmental factor. Basing on the probability of the target location and the shared data, the distributed probability map model is implemented.

Authors in [209] proposed a real-time system for vehicle detection using UAVs in rural environments. Firstly, the fast detection solution is applied which rejects the background and selects patterns of the man-made objects. Then, the selected patterns are used in the classification techniques. They compared their algorithm to many feature extraction methods (histogram of gradients and Gabor coefficients) which are applied for this specific task and they considered four classification techniques (KNN, SVM, decision trees and random trees). In [55] authors proposed a framework for heterogeneous UAVs networks to detect fire. They integrated the infrared sensors and the data fusion elicited from computer-aided modules. The localization of fire detection is based on the UAVs position and the localization of cameras.

In [210] authors proposed a target detection and localization mechanism using a team of cooperative UAVs and UGVs. They applied a customized motion detection algorithm to follow the team from the moving camera equipped on the UAV. The UAVs localization algorithm converts the teams image locations to their real-world positions, basing on perspective transformation. They used also a rule-of-thumb localization method to estimate the detected individuals geographic locations. Experimental results show the effectiveness of this framework for autonomous surveillance. Once the targets are detected, it will be tracked by the UAVs swarms. Therefore, tracking can be implemented in the object detection application. In [208] authors studied the high-level definition of the cooperative tracking system. They also described some elements (vehicles, the tracked object, lock guidance, and geo-location) by giving the technical details.

In [211] authors presented an example of a search and tracking application. The proposed application switches between the search and tracking algorithms and use the information gathered during past operation. Moreover, the application treated the scenario of the tracked target loss. In this case, the search algorithm is executed again by using recursive Bayesian filtering. In [212] authors are based on the vision measurements to build the collaborative system. The gathered information is fused to minimize the transmitted data between vehicles. Cameras are orientated to POI (Point Of Interest) with some adjustments according to the UAV and target movement.

11.7. Healthcare

UAV equipped with artificial intelligence systems and sensors have been used in various humanitarian efforts to transfer small aid packages or to gather vital information to victims affected by natural disasters or infectious diseases. In the following, we focus on the proposed research work for the UAV exploitation in the current pandemic COVID-19. Authors of [213] designed a UAV system to detect the coronavirus from the thermal image with less human interactions. The proposed system used a Virtual Reality screen with live video monitoring to command the camera for scanning people.

Moreover, UAVs can be used to deliver goods in the highly trafficked areas to increase social distancing, decrease person-to-person contact, and thus reduce the spread of COVID-19.

11.8. Delivery and transportation

Delivery systems based on UAV are increasingly being exploited in the logistics field, especially to achieve rapid last-mile delivery. For example, authors of [214] proposed a prototype of an autonomous drone for packet delivery. UAVs must find the right building using GPS and drop the packet at the right balcony. The proposed system can be profitable, however, its existence in a realistic urban environment does need some requirements.

In [215] UAVs are incorporated into vehicular networks for smart transportation networks. UAVs act as a relay to deliver data packets between ground vehicles. The data forwarding rules are based on the estimated delivery delay and the data packets' delivery probability. The proposed scheme depends on the calculated ground vehicle mobility which can be so far to the reality, therefore, it is required to perform a vehicle mobility prediction algorithm.

Taxi-UAV is one of the most important technologies for future smart cities to reduce congestion and traffic. The urban air transportation can alleviate the transportation congestion of the ground and enable rapid, reliable transportation within cities.

12. UAVs network design tools

In this section, we study the existing projects, test beds and simulators of the UAVs networks [216] to offer a quick guideline to the new researchers. Authors in [217] developed a novel multi-UAV system basing on wireless ad-hoc networking architecture. The proposed system validated successfully the feasibility and the effectiveness of wireless ad hoc networking using UAVs. Moreover, there a wide variety of testbeds which are available to be used by scientists. Researchers in the Aerospace Controls Laboratory at MIT utilize a UAV testbed, Real-time indoor Autonomous Vehicle test Environment (RAVEN) [218]. RAVEN exploits a motion capture mechanism to facilitate the prototyping of the flight controllers for airplanes and helicopters, vision-based sensing for indoor flight and robust coordination for multiple helicopters. Another indoor testbed is developed at the University of Pennsylvania named General Robotics Automation Sensing and Perception (GRASP) [219]. The proposed testbed support research on a dynamic and coordinated flight of UAVs.

On the other hand, there are also wide variety of UAVs simulators such as X-Plane [220] and Microsofts Flight Simulator [221] which are commercially available to simulate UAVs networks. X-Plane [220] is developed by Laminar Research. It received FAA certification and it is used with some hardware configurations to increase its fidelity to the flight model. The Microsoft Flight Simulator [221] gives detailed visuals for the UAVs and environment. It uses a less accurate search table to operate the flight model of the UAVs simulation. Moreover, there are some other available open-source UAVs simulators such as FlightGear [222]. FlightGear uses a blade element analysis by default, similar to X-Plane. UAVs models must be created by an external 3D modeling application with an XML file illustrating the various UAVs features. FlightGear is used for various academic projects.

13. Open issues and recommendations

13.1. Open issues

Although many types of research have been performed to increase the efficiency of UAVs networks, there are many unsolved problems that should be investigated in future works. In the following, we present several issues concerning system integration and interaction of the multi-UAV system components. As mentioned, the multi-UAV systems design is application-dependent. We focus on open issues of our current

research which is cooperative target tracking using multiple UAVs, especially from the remote sensing and coordination viewpoints.

UAV platform: The UAVs flight dynamics platforms (e.g., sensitivity to weather and wind) as well as the UAVs orientation and position have a high influence on the communication links.

Data processing: UAVs already are equipped with several sensors providing information. Data processing requires a great computational power that might not be possible onboard UAVs. Cloud computing may provide a solution to this issue.

Routing: In the UAVs networks, the network topology can change frequently. Data routing between UAVs is a serious challenge. The routing protocols must update routing tables dynamically basing on the topology changes.

Path Planning: In multi-UAV operation and mission area, co-operation between UAVs is a crucial feature to increase system efficiency. Frequently, each UAV must modify its previous path, and then it will re-calculate dynamically the new ones. Therefore, new algorithms for dynamic path planning are needed to ensure coordination between UAVs.

QoS provisioning: Nowadays, UAVs are used in many types of applications, so it can transport different categories of data such as GPS locations, simple text messages streaming video/voice. Therefore, the UAV network needs to support the QoS to satisfy some predetermined service performance constraints including packet loss rate, delay, bandwidth, etc.

UAV mobility and placement: In the mission area, the UAVs can use different sensors (one UAV can be equipped with an infrared camera; another UAV is loaded with a high-resolution camera). Therefore, multiple images are transferred in the same area, so there exists an open issue to optimize the UAV mobility and placement to decrease energy consumption while diffusing data.

Handovers techniques: Some applications like surveillance large areas require long-range communication and it must maintain connectivity between intermediate infrastructure depots. In such applications, handover techniques would be required.

Integration of UAVs into existing systems: Multi-UAV systems can be a part of the existing infrastructure. Therefore, although UAVs have their own features, they need to take into account the challenges associated with the larger systems for performing efficient integration and operation, like scalability, increased data volume, and data processing from a set of heterogeneous devices. For this reason, the techniques of integrating UAVs into a large-scale network are an emergent challenge.

13.2. Recommendations and further work

UAVs have prodigious potential to revolutionize the future of several fields and sectors by acting as wireless relays to expand connectivity within ground networks. In this context, in the following, we highlight some recommendations and further works related to several issues.

- It is recommended that UAV sensors, networks, processes, technologies, and systems should provide new ways for military, civilian, commercial, and public applications to offer less stressful and safer environments, reduce the humanitarian risk exposure, enhance seamless access, and improve the global connectivity. A key goal is to develop multi-UAVs systems that should maintain reliable communication, remotely control and command with unlimited operation range, securely transmit information, and support the co-existent mobile broadband applications. The aforementioned recommendations should attract the attention of the academic community to develop a set of new international standards to

optimize operating multi-UAVs systems

- Besides that, it is recommended that the network infrastructures of the Internet of Medical Things will include Wireless Body Area Networks (WBANs), clouds or remote servers, and UAVs. UAV will be deployed to gather patients information from multiple WBANs and to send the collected information to cloud servers or medical servers via cellular nodes (5G or 6G base station) providing the required connectivity. However, several challenges and further research work may be addressed in order to improve UAV-based data gathering in advanced healthcare systems. Existing work and contributions related to UAVs routing mechanisms are not competent, most of them do not cogitate the issues of UAVs' high speed mobility, network load balancing of UAVs. Besides that, the dynamic aspect of UAV leads to the necessity of multiple UAVs to cover a particular area for data gathering which requires advanced mechanisms to address the coverage issues in the context of smart healthcare systems under for synchronization and collision avoidance constraints. Therefore, further investigations are required to address the aforementioned issues.
- Moreover, the UAVs ability to get real-time, high-resolution spatial and temporal information at low cost makes them potential technology for epidemiology research, disaster management, and emergency medicine. In this context, We must accelerate research efforts correlated to UAVs integration in the medicine domain with a focus on, safety, participation expansion, response time, security, privacy, and the development of new dedicated applications and services.
- There are different types of UAVs using a widespread variety of technologies. The lack of standardized monitoring procedures prohibit effective UAVs activities. The identification of a specific UAV and its associated pilot and owner are of greatest concern to regulatory authorities. To achieve this goal, it is recommended to associate object identifiers to UAV devices with all stages of the UAV life-cycle (manufacturing, marketing, repairing, and scrapping) and with the owner or with the user personal information through regular registration procedures. Regarding this open issue, further work could be done regarding specifications of registration procedures and assignment rules of object identifiers used for UAVs.
- Real experiments with UAVs are fundamentally difficult due to hard regulations and the necessity of large open space. Thus, the majority of researchers resort to simulations as an alternative to evaluate and to analyze the performance of their suggested systems, protocols, and mechanisms. However, simulations assume that UAVs can move in any time at any direction without any explicit constraints and restrictions related to obstacles or used hardware or weather conditions. Consequently, such simulation results may be so far from realistic results. It would be useful to develop available dedicated UAVs simulators allowing researchers and developers to accurately simulate different types of UAVs according to their diverse constraints.
- Furthermore, it is recommended that 5G technologies, processes, and systems should unveil a sophisticated novel mobile communications network architecture that not only improves seamless access and physical data rate, but also creates a new innovative ecosystem allowing the deployment of pioneering management, services, and applications. A key target is to build a fully-fledged innovative network architecture that should support not only classical mobile broadband applications and services but also a stream of vertical sectors such as UAV based intelligent transport, secure IoT and cloud computing, eHealth, non-intrusive AI-based automated management and user assistants and other yet to be defined 5G-based services." Therefore, for UAVs as a vertical sector associated with 5G, The third generation partnership group (3GPP)

defined UAVs as a study item with four fundamental objectives: (1) Understanding UAV traffic requirements; (2) Development of a channel model to characterize air-to-ground propagation characteristics; (3) Determination of whether the current LTE infrastructure could be reused to provide cellular service to UAVs; (4) Definition of the required enhancements to effectively serve UAVs.

- The International Telecommunication Union Telecommunication Standardization Sector (ITU-T) defined the work item Y.UAV-BSI for providing requirements and functional architecture of base station inspection services using UAVs.
- The American National Standards Institute (ANSI) defined standardization Roadmap [223] for UAV systems assessing gaps, and making recommendations for priority areas where there is a perceived need for additional standardization effort related to UAVs research and development. The roadmap has identified 60 gaps (3 gaps identified as a low priority, 17 as medium priority, and 40 as a high priority), examined 64 issue areas, and defined recommendations across the topical areas of flight operations, airworthiness, commercial services, public safety operations. A gap means there is no specification or published standard that considers the issue. Therefore, the roadmap is a good reference for the future development of standards for UAVs systems that identify emerging issues necessitating further elaboration.
- The Institute of Electrical and Electronics Engineers (IEEE) standardization body working on several future UAVs standards (started on March 2019 and are now active):

- (1) IEEE P1920.1: Standards for Aerial Communications and Networks [224] that will define air-to-air communications for self-organized ad hoc aerial networks.
- (2) After achieving IEEE P1920.1 standard, it is expected to start the activities related to the IEEE P1920.2 [225]: Standards for Vehicle-to-Vehicle Communications for UAS (Unmanned Aircraft Systems).
- (3) IEEE P1936.1 Standard for Drone Applications Framework (SDAF) [226] that aims to establish a framework for drone applications support and to specify drone application classes and scenarios and the required application execution environments.
- (4) IEEE P1939.1 [227] project aims to define a standard for a framework for structuring low altitude airspace for UAV. It defines UAV capabilities and UAVs related infrastructure to operate in and comply with low altitude air space regulations.
- (5) IEEE P1937.1 (specifies standard interface requirements and performance characteristics of payload devices in drones (IPDD) [228]. It expresses the interfaces, provisioning, performance metrics, operation control, and management for UAV payload devices.

The above recommendations would require a series of new international norms and defacto standards, and seamless transparent interfaces and processes to ensure user-geniality, stretching well into amiability and affordability.

14. Conclusion

UAVs play a prominent role in the operation area, the complicated missions and the broader range of applications. UAVs should cooperate to perform complex tasks particularly in inaccessible areas from the ground, and it is required to quickly and easily organize a networked system, hence the need to form a multi -UAV networks. In such type, the control algorithms and transmission range are very important. UAVs ad hoc formation can solve various problems of surveillance, reconnaissance, and navigation. In this paper, we discussed the UAVs systems considerations including mobility, adaptability, reliability, scalability, latency, UAV platform constraints, bandwidth allocation,

power consumption, and computational power, localization and security and privacy. We provided also a review of the existing UAV's communication protocols and mechanisms. We presented current research in the area monitoring with UAVs. Moreover, we presented the existing multi-UAV projects, testbeds, and simulation environments. Furthermore, we discussed open research issues for UAVs networks. We hope that this paper will be useful for many researchers and professionals interested in the design and implementation of UAV protocols, network infrastructures and applications.

Declaration of Competing Interest

On behalf of all the authors, I hereby declare there is no conflict of interest with any other party whatsoever.

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